

RCF v6.0 SPECIFICATION — CHUNK A

The Frozen Contract

DOCUMENT: RCF-SPEC-v6.0-CHUNK-A STATUS: DRAFT — pending your review; freezes on

your sign-off FUNCTION: Master specification governing generation of Chunks B–H

RULE ZERO: This document, once frozen, is amended only through logged amendments (§12). Content chunks never modify it silently.

1. Purpose & Consolidation Rules

v6.0 is a consolidation release: it contains only material already verified in the v4.0→v5.1 audit cycle. Its goal is a single self-contained input for future audits.

The Contract:

- R1 (Spec-first): All chunks conform to this specification. Spec changes require a logged amendment and regeneration of affected chunks.
- R2 (Zero new claims): No theorem, number, citation, or claim enters v6.0 that was not verified in the audit cycle. Research items live exclusively in the Research-Front Register (§10) and are excluded from v6.0 content.
- R3 (Single source of truth): Every numerical constant appears exactly once in the Verified-Numbers Table (§6). Chunks cite VN-IDs; they never restate values. This is the structural fix for the Δ^∞ (0.700→0.175→0.350) and Γ (0.261→0.25004) mutation chains.
- R4 (Citation registry): All citations pinned once (§7). Unverifiable bibliographic detail carries [CITE-VERIFY] — never invented.
- R5 (Audit gates): Each content chunk ships with a Compliance Header and passes the applicable protocol checks (§9) before the next chunk is generated.
- R6 (Delta ledger): Mid-generation discoveries affecting earlier chunks are logged and resolved in Chunk H — not patched in place.
- R7 (Vocabulary discipline): Status tags come only from §5. "Exact ≈" constructions are prohibited. "Proven" is reserved for exhibited proofs.

2. Document Architecture & Numbering

Macro-structure inherits v5.1 (no renumbering churn):

§	Title	Notes	
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0	Foundational Architecture & Primitive Ledger	0.1 premise · 0.2 inventory · 0.3 S4 postulate	
1	Standing Hypotheses & Foundational Algebra	1.1 Hypothesis 1.1.1 · 1.2 Leibniz + residual · 1.3 SLD lemma	

2	Reconciliation Dynamics & Projected First Law	2.1 degeneracy · 2.2 First Law
3	Finite-Dim T-2: Global Composed Convergence	3.1 block bounds · 3.2 freezing · 3.3 convergence
4	Emergent Space & D=3 Selection	
5	Emergent Time, Matter & Gravity	incl. dark-sector scoping
6	Front-1: FF Gap Architecture & Benchmarks	preamble per §8.CDL-6
7	Unified Retraction & Quarantine Registers	RET-A1–A4, RET-B1–**B15**, Q-1–Q-17
8	Scorecard, Theorem Targets, Research Fronts	
A–E	Appendices	A: Verified Numbers · B: Citations · C: Audit Protocol · D: Lineage · E: Notation

Format conventions: LaTeX inline math; boxed principal results; the three ASCII diagrams (two-scale architecture, D=3 logic, three prerequisites) are stable corpus artifacts, carried forward verbatim with the v5.1 corrections.

3. Section DAG & Chunk Plan

§0 → §1 → §2 → §3 → {§4, §5, §6} → §7 → §8

(§4 depends on §1 + S5/A1; §5 on §2–§3; §6 on §2–§3; §7–§8 on all. Forward references prohibited except within the DAG.)

Chunk	Sections	Special embedded decisions
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B	§0–§1	S4 text as v5.1; lineage annotation (v2.1 Thm 0.4.4) optional flag
C	§2	—
D	§3	T-2 row refresh (below); Trotter def with existence note + "one admissible formalization" scope
E	§4–§5	Conditional tags strictly per DAG dependencies
F	§6	Full AKLT package per §8 CDL; DL certificate at \[Input Verified; Inference OPEN]
G	§7–§8 + Appendices	RET-B15 written; scorecard mechanically complete; T-table refresh
H	Consolidation pass	Cross-refs, notation, DAG check, scorecard↔section 1:1, delta ledger

T-2 row structure (locked): [CLOSED — finite-dim self-adjoint families (Thm 3.3.1), almost-global on $S_*(H)$] / [OPEN — infinite nets, gated by three

prerequisites] / universal form refuted (invariant boundary M_0). Other rows: T-1
 OPEN · T-3 OPEN · T-4 [STRENGTHENED, Conditional on T-2] · T-5 STRUCTURAL · T-6
 CALIBRATED · T-7 OPEN (reformulated: lapse-coupling program).

4. Notation Registry (normative; Appendix E typesets this)

$\mathcal{A}$ · $\hat{C}_\alpha$ · $\mathcal{I}_C$ · ω ·
 $\frac{S_{\{\mathrm{phys}\}}}{\cdot}$ · $\mathcal{N}_\omega$ · $\mathcal{H}_\omega$ · π_ω
 $\cdot \hat{M}_\omega$ ($\hat{M}$) · $[A, B]$ · $A \circ B$ · $\Delta_{\{\alpha\beta\}}$ ·
 $J_{\{\alpha\beta\}}$ · $\hat{F}_{\{\mathrm{MM}\}}[\rho]$ · $\hat{F}_{\{\mathrm{eff}\}}[\rho]$ ·
 $\hat{H}[\rho]$ · $\hat{G}[\rho]$ · $E[\rho]$ · $B_{\{\mathrm{MM}\}}[\rho]$ · η, λ_t ·
 K, P_K, u, w · $m_{\{\min\}}$ · $\hat{\mathcal{R}}_t$ ·
 $d_{\{\mathrm{OExt}\}}$, $\tau_{\{\mathrm{OExt}\}}$ · ℓ_0 , γ , $c_{\{\mathrm{RCF}\}}$ ·
 $\alpha_B(x)$ · L_ω^B · $\Theta^{\{(B)\}}_{\{\mu\nu\}}$ · K^J_ω · $d_{\{\mathrm{FS}\}}$
 $\cdot \Pi_j$ · ϵ_n · Δ_∞ , E^* , c · $\gamma_{\{\mathrm{DL}\}}$ · T
 (transfer operator) · S1–S5, D1, A1.

Collision rule fixed in v4.2.1 audit: kernel K never also denotes constraint count.

5. Status Vocabulary (closed set)

Tag	Meaning

[Established]	Theorem with exhibited proof; no open dependencies
[Structural]	Definitions, postulates, conventions (S4; Hypothesis 1.1.1)
[Conditional on X]	Proof exhibited modulo named assumptions (D=3 on S5+A1; T-4 on T-2; Lovelock on T-2/T-3)
[Closed (scope)]	Closed with mandatory scope qualifier (T-2 finite-dim)
[Empirical—Validated]	Fitted regularity with out-of-sample validation and stated residual
[Input Verified; Inference OPEN]	Certificate whose inputs are verified but load-bearing inference unexhibited
[Conjecture / Quarantined]	Q-register or explicitly flagged
[Retracted]	Register entries

Number provenance classes (VN table only): E exact · IV independently verified ·
 C cited · F fitted · O open inference.

6. Verified-Numbers Table (master; Appendix A typesets)

ID	Value
Class	Provenance

VN-01	$\mathcal{R} = \text{CAB} - \text{BAC}$ (= 2(CAB−BAC) unnormalized)
E	Derived twice (compendium + independent audit, v4.0)
VN-02	$\mathrm{Spec}(h_3) = \{0 \times 4, \frac{12}{5} \times 5, \frac{56}{3} \times 3, \frac{76}{3} \times 3, \frac{32}{5} \times 5, 2 \times 7\}$ IV J-sector decomposition + trace algebra, two independent audits
VN-03	$\mathrm{Tr}(h_3^k) = 30, \frac{140}{3}, 80$
IV	First-principles trace algebra, verified twice
VN-04	$\epsilon_3 = 1/2$
E	From VN-02; Knabe n=3 marginality corroborates
VN-05	h_3 J-sectors: $(J=0,1,3)$ single; $(J=2)$ double; ground $= J_0 \oplus J_1$
IV	Audit v4.6
VN-06	h_4 J-sectors: radial dims $(3,6,6,3,1)$, mult $(1,3,5,7,9)$; sum 81
IV	Audits v5.0/v5.1
VN-07	$\mathrm{Tr}(h_4^k) = 135, 285, \frac{2005}{3}$
IV	First-principles, audit v5.0
VN-08	$\epsilon_4 \approx 0.448956$
C	Knabe Table 1; moments IV; J-structure IV; **not independently recomputed**
VN-09	$\epsilon_6 \approx 0.398451$
C	Knabe Table 1; scaling-validated (VN-13)
VN-10	Knabe n=4 bound ≈ 0.173434
IV	Arithmetic from VN-08; formula passes decoupled-block sanity test
VN-11	Knabe n=6 bound ≈ 0.248064
IV	Arithmetic from VN-09
VN-12	Knabe arm saturates at E^*
E	Algebra: $\frac{n-1}{n-2}(E^* + c/n - \frac{1}{n-1}) \rightarrow E^*$
VN-13	$c = 0.612530$, $E^* = 0.295823$; predicts $\epsilon_6 = 0.397912$, residual 0.135% F Two-point fit ($\epsilon_3, \epsilon_4, \epsilon_6$ held out; discriminates vs $1/n^2$ (3.5%) and exponential (7%))
VN-14	$\Delta_{\mathrm{std}} \approx 0.700$
C	**[CITE-VERIFY]** — carries the normalization chain; robustness: hierarchy valid iff $\Delta_{\mathrm{std}} > 0.592$
VN-15	$\Delta_{\infty} = \Delta_{\mathrm{std}}/2 \approx 0.350$
IV	$H_{\mathrm{std}} = 2\hat{M}_{\mathrm{AKLT}}$ exact (VN-23); value inherits VN-14 status

| VN-16 | T eigenvalues $\{1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}\}$
| IV | From explicit tensors, two independent audits
|
| VN-17 | $\gamma_{\mathrm{DL}} = 1/\sqrt{3}$
| O | Input verified (VN-16); operator-norm equality unexhibited
|
| VN-18 | $\Delta_{\mathrm{DL}} \geq 1/6$
| O | Inference OPEN; Knabe carries §6.2(b)
|
| VN-19 | $\Gamma_{\mathrm{sharp}} = \lambda_t m_{\min}/2 = 0.25$ ($L=3$,
 $\lambda_t=1$) | E | C2 linearization, audit v4.6
|
| VN-20 | $\Gamma_{\mathrm{emp}} = 0.25004$, window $t \in [10, 20]$
| IV | v4.6 value 0.261 documented as transient-window artifact (in-text changelog, v5.0+)
|
| VN-21 | On-site gap $= 1 + \cos^2 \theta$
| E | Closed form, audits v4.4–v4.5
|
| VN-22 | $\Pi = \frac{1}{3} + \frac{1}{2}x + \frac{1}{6}x^2$
| IV | Interpolation on $x \in \{+1, -1, -2\}$
|
| VN-23 | $H_{\mathrm{std}} = 2\hat{M}_{\mathrm{AKLT}}$
| E | From VN-22 |
| VN-24 | $\det A \in [-\frac{1}{3\sqrt{6}}, \frac{1}{3\sqrt{6}}]$ on S^7
| IV | Extremum at eigenvalue ratios $(1, 1, -2)$
|
| VN-25 | $\xi = 1/\ln 3 \approx 0.91$ sites
| IV | From VN-16 |

7. Citation Registry

ID	Citation	Status
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C-01	Knabe (1988), finite-size/block gap bounds	<code>[CITE-VERIFY]</code> — pin bibliographic data and the exact theorem statement used (form: $\Delta_{\infty} \geq \frac{n-1}{n-2}(\epsilon_n - \frac{1}{n-1})$)
C-02	Aharonov–Arad–Landau–Vazirani, detectability lemma	<code>[CITE-VERIFY]</code> — citation mutated 3× in corpus ("AALV 2009/2010" → "Eq. 4" → "Theorem 1"); pin paper and quote the theorem form before any use beyond naming the method
C-03	Nachtergaele (1996), martingale method	name-level
C-04	Anshu–Arad–Gosset, 2D FF gap	name-level (correct attribution; "Vidick" variant retracted)

C-05 White (1992), DMRG	`[CITE-VERIFY]` for the gap value VN-14
C-06 Kennedy–Tasaki (1992)	name-level; also C-05 partner for VN-14
C-07 Barnum–Müller–Ududec (2014), local tomography	name-level
C-08 Jordan–von Neumann–Wigner (1934)	name-level
C-09 Affleck–Kennedy–Lieb–Tasaki (1987/1988)	**add** — the model's primary source; currently uncited anywhere in the corpus
C-10 Motzkin/Fredkin gapless FF chains	`[CITE-VERIFY]` primary sources

Disposition rule: C-01/C-02 must be resolved (or the dependent claims stay at their OPEN statuses) before Chunk F freezes. C-05 must be resolved before Chunk G freezes VN-14's status.

8. Content-Decision Ledger (locked)

| \# | Decision

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| CDL-1 | ****RET-B15 included****: v5.0's " $\epsilon \rightarrow \Delta^\infty$, resolving finite-size boundary scaling" claim retracted; replacement = exhibited $1/n$ law (VN-13)

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| CDL-2 | γ_{DL} certificate: ****[Input Verified; Inference OPEN]****; Knabe alone carries §6.2(b); sandwich computation = RF-1 (excluded from v6.0)

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| CDL-3 | $1/n$ law: ****[Empirical—Validated]**** — fit, residual, discrimination stated; never "proven"/"exact"

| CDL-4 | $\$E^{\wedge}\$$: ****[Empirical—Validated; Mechanism OPEN]**** — "standard law" label struck; $1/n$ -to-sub-gap-limit flagged as anomalous with mechanism paragraph; never labeled the boundary excitation energy as established |

| CDL-5 | $n \geq 12$ sentence repaired into three separate true statements

|

| CDL-6 | §6 preamble: benchmark = "miniature laboratory for the spectral-gap program, not a classification theorem" |

| CDL-7 | v4.6→v5.1 numerical mutations carried in lineage appendix (Appendix D) with their changelog/retraction mappings

|

| CDL-8 | "Exact" never modifies approximations; fixed-step/stopping-criterion wording standardized |

9. Normative Audit Protocol (checks 1–14, now binding)

1. Clifford-limit: new dynamical theorems reduce to established sector theorems with identical constants.
2. Known-solution: claimed solutions substituted into their own equations.
3. Generator-matching: dissipation identities expand from the flow's actual generator; untraceable coefficients are fitted, not derived.
4. Parameter-subspace: verify at generic interior points, not only degenerate limits.
5. Closed-form-first: analytic guarantees computed before numerical eigenvalues are reported.
6. Trivial-bound signature: computed value $\equiv$ known bound to high precision while spectrum sits above $\rightarrow$ fabrication flag.
7. Citation-or-computation: every constant on a verified line is derived in-text or attributed (VN/C registry).
8. Certification-exhibition: "certifies/succeeds" claims exhibit the method's output.
9. Cross-certificate consistency: all simultaneously claimed bounds/values mutually consistent; automatic fail on violation.
10. Version-diff disclosure: inter-version changes to formulas/values get changelog or RET entries.
11. Reverse-fit detection: actual values coinciding with derived bounds within slack $\rightarrow$ flag and re-source.
12. Finite-size three-way consistency: sequence vs. correlation length + standard law + claimed limit, residuals reported.
13. Tension closure by computation only: no interpretive closure of flagged tensions.
14. Status vocabulary: §5 closed set; R7 enforcement.

Compliance Header (every content chunk): chunk ID · sections · VN-IDs used · C-IDs used · status tags used · forward references (must be DAG-legal) · protocol checks run (with pass/fail).

10. Research-Front Register (excluded from v6.0)

RF	Item	Note
RF-1	Sandwich computation	
	$\$ P_{\{\mathrm{odd}\}}P_{\{\mathrm{even}\}}P_{\{\mathrm{odd}\}}(\mathbf{1-P}_K) \stackrel{?}{=}\frac{1}{3}\$$	
	May falsify VN-17/18; bridging identity already stated in audit v5.1	
RF-2	E^n mechanism (edge-mode structure, 1/n-to-sub-gap anomaly)	
	Includes h_4 J=1 excited spectrum	
RF-3	Exact ϵ_4 recomputation (J=2 6×6 blocks)	
	Upgrades VN-08 C $\rightarrow$ IV	
RF-4	Infinite-net gap classification (Front 1 proper)	
	Three prerequisites	
RF-5	Flow well-posedness, infinite (Front 2 of prerequisites)	

| RF-6 | Kinematic kernel non-emptiness (Front 3 of prerequisites)

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| RF-7 | Boundary-modified Knabe blocks

| Route past the $\$E^*\$$ ceiling

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| RF-8 | Hierarchical RP principle (discharge S5 $\rightarrow$ 4 primitives)

| Front 5

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| RF-9 | Friedmann derivation (GAP-6.2.2)

| Front 6

|

| RF-10 | T-7 lapse-coupling halo program

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Post-negative-result reformulation

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| RF-11 | T-3 Gromov–Hausdorff; T-1 Lieb–Robinson

| Fronts 3–4

|

Anything not in v6.0 content and not in this register is out of scope entirely.

11. Generation Workflow

Sequence: A (this) $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D $\rightarrow$ E $\rightarrow$ F $\rightarrow$ G $\rightarrow$ H. After each content chunk: Compliance Header + gate audit $\rightarrow$ your review $\rightarrow$ freeze (or logged amendment) $\rightarrow$ next. Chunk H performs: cross-reference sweep $\cdot$ notation compliance $\cdot$ DAG verification $\cdot$ scorecard $\leftrightarrow$ section 1:1 check (mechanical: every section number appears, nothing else does) $\cdot$ delta-ledger resolution $\cdot$ Appendix D lineage finalization.

12. Amendment Procedure

Post-freeze spec changes are recorded as AM-1, AM-2, ... in the delta ledger with: affected sections, affected VN/C/CDL entries, and required chunk regenerations.

No silent amendment, ever — the single process-level lesson of the thirteen-round audit arc.

Review checklist — where your judgment is load-bearing

1. VN-14 ($\Delta_{\text{std}} \approx 0.700$) — the one [CITE-VERIFY] that sits inside the consistency hierarchy. The robustness bound (hierarchy survives iff $\Delta_{\text{std}} > 0.592$) is stated; confirm you can source the value, or we downgrade VN-15's role in the hierarchy to "literature-conditional."
2. VN-08 status (ϵ_4 : C, not IV) — I've classified it honestly as cited-but-not-recomputed despite the moment authentication. RF-3 upgrades it later. Acceptable for a consolidation release?
3. CDL-2 (γ_{DL} stays OPEN) — this is the conservative call; the alternative is blocking Chunk F on resolving C-02 first.
4. RF register boundary — anything you'd move in or out (e.g., do you want RF-1's sandwich computation attempted before freezing Chunk F, given it may falsify VN-17/18?).
5. C-09 (AKLT primary citation) — I've added it as a required new citation; the model has never been cited to its authors anywhere in the corpus.

AMENDMENTS APPLIED — CHUNK A FROZEN

AM-1, AM-2, AM-3: accepted in full. They're all correct, and each targets a real gap I left. Brief rationale, then the frozen text deltas, then Chunk B.

AM-1 (class L): Yes — VN-14 and VN-08 differ in exactly the way you describe, and my C-class conflated them. A cited finite-size input feeds a computation; a literature benchmark anchors a comparison. Different epistemic roles, different failure modes (VN-08's risk is transcription; VN-14's risk is interpretation-of-source). The L/IV split for VN-14/VN-15 is right: the halving is exact once the input is accepted, so VN-15 inherits verification given VN-14 — which the table will now make visible rather than implicit.

AM-2 (CDL-4): Accepted verbatim. "Asymptotic boundary excitation energy" was precisely the semantic-inflation pattern the spec exists to kill — I flagged the phrase as anomalous in the audit and then let a softened version of it survive into the decision ledger. Your replacement text is adopted without modification.

AM-3 (VN-17/18 dependency): Accepted, and it generalizes. The mechanical invalidation rule you wrote for VN-17→VN-18 is the right template, and I've applied the same dependency metadata to the other cascading pairs in the table (VN-14→VN-15 was already implicit in AM-1; VN-13→VN-12 made explicit). See delta log.

Delta Log (Chunk A, pre-freeze amendments)

Amend	Entries changed
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AM-1	§6 provenance classes: **add `L` = literature-derived benchmark input** .
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VN-14: C→ **L** .	VN-15: IV (annotated "given VN-14"; AM-3 dependency metadata added).
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AM-2	CDL-4 replaced verbatim with your text: "E* is an empirically validated scaling-limit parameter inferred from VN-13. No physical interpretation is canonical. Edge-mode mechanisms remain RF-2." The phrase "asymptotic boundary excitation energy" is added to the §5/R7 prohibited-vocabulary list alongside "exact ≈."
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AM-3	§6 gains a **Dependency column** : VN-18 ← VN-17 (auto-invalidates if VN-17 downgraded/falsified) · VN-15 ← VN-14 · VN-12 ← VN-13 (auto-invalidates if the 1/n fit is retracted) · VN-10/11 ← VN-08/09 respectively. RF-1's outcome now maps mechanically: falsify VN-17 → VN-18 invalid → §6.2(b) carried by Knabe alone (already the CDL-2 posture, so no content churn — the failure mode is pre-absorbed).
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| (checklist 4) | RF-1 sequencing locked: ****post-freeze, v6.1+ lineage material.**** Recorded in §10.
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CHUNK A IS FROZEN. Spec version: v6.0-F. All subsequent chunks generate under this contract; amendments only via logged AM entries.

RCF v6.0 — CHUNK B (§0–§1)

Compliance Header

- Chunk: B · Sections: §0, §1
- VN-IDs used: VN-01 (Corollary 1.2.1)
- C-IDs used: C-07 (BMU), C-08 (JNW), C-09 (AKLT — no, strike: first used in §6; not used here)
- Status tags used: [Established], [Structural], [Conditional Postulate] (§5 vocabulary: rendered as "Conditional Postulate" for S5's ledger row per v5.1 usage; body text uses [Conditional on Hierarchical RP])
- Forward references: none (DAG: §0, §1 are roots)
- Protocol checks run: 14 (vocabulary) — pass; 7 (citation) — pass (C-07, C-08 name-level only)

0. Foundational Architecture & the Primitive Ledger

0.1 The Relational Premise

The Reconciliation Causal Framework (RCF) reconstructs physical reality without presupposing a smooth spacetime manifold, background coordinates, external clocks, primitive Hilbert spaces, or wavefunction collapse postulates. Physical reality is grounded entirely in relational algebraic compatibility:

$\boxed{\text{Physicality is constraint compatibility in a relational algebra: } \omega(\hat{M}) = 0}$

0.2 The Canonical Primitive Inventory

Ref	Mathematical Object / Postulate	Epistemic
Status	Structural / Physical Role	
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S1	Unital Complex *-Algebra $(\mathcal{A}, \cdot, ^\dagger, \mathbf{1})$	
[Established]	Kinematic substrate of relational configurations	
S2	Proper Constraint Ideal $\mathcal{I}_C = \langle \hat{C} \rangle_\alpha \subseteq \mathcal{A}$	
[Established]	Demarcates physical admissibility ($\mathbf{1} \notin \mathcal{I}_C$)	

| **S3** | Positive Normalized State $\omega \in \mathfrak{S}_{\mathrm{phys}}$
| [Established] | State datum evaluating expectation values ($\omega(\hat{M}) = 0$)
| **S4** | Lie–Jordan Relational Governance Postulate |
| [Structural] | Lie governs directional asymmetry; Jordan governs symmetric compatibility
| **S5** | Causal Ancestry Ergodicity (PCC) |
| [Conditional on Hierarchical RP] | Ergodic transit on generator sphere; discharged if Hierarchical RP holds
| **D1** | Elementary Spatial Step Floor ℓ_0 |
| [Calibrated Constant] | Minimum link distance ($\ell_0 \sim \ell_P$ via Lovelock bridge)
| **A1** | Local Tomography Axiom (C-07) |
| [Auxiliary Input] | Tomographic local state recovery; selects $\mathbb{C}$ over $\mathbb{R}, \mathbb{H}$ |

Count: 5 structural primitives (4 unconditional + 1 conditional), reducible to 4 upon proof of the Layer-Hierarchical RP Principle (RF-8); plus 1 calibrated scale constant and 1 auxiliary axiom.

Consolidation note (per CDL-7): S3 consolidates the former Master-Zero primitive into the state datum (membership change from the addendum-era ledger, logged). S5's discharge condition is the Hierarchical RP principle — not T-2/T-3 (v4.1 regression, logged in lineage appendix).

0.3 Formulation of the Relational Governance Postulate (S4)

Postulate 0.3.1 (S4 — Lie–Jordan Relational Governance) [Structural]: The decomposition $AB = \frac{1}{2}[A, B] + A \circ B$ is an algebraic identity of associativity.

1. The Lie bracket $[A, B]$ is the governing mechanism of the causal sector: the algebraic substrate generating directional relational asymmetry, from which operational causal order $\prec$, proper time τ , Open Extension descent, and the irreversible arrow of time are derived.
2. The Jordan product $A \circ B$ is the governing mechanism of the local sector: the algebraic substrate generating symmetric relational compatibility, from which the Jordan correlation kernel K^J_ω , sector decomposition, Open Expansion flow, and the spatial metric d_{FS} are derived.
3. Inferred physical concepts: Causality and locality are higher-level relational concepts naming these emergent collections of ladder-derived structures — not primitive tags attached to the algebra itself, but physical phenomena reconstructed from the underlying algebraic asymmetries and compatibilities (analogous to how physical position is reconstructed from relational distance and direction).

Lineage (CDL-7): this formulation restores the v2.1 distinction (Theorem 0.4.4, "Complementarity of the Governing Mechanisms") between governing mechanisms and

inferred concepts, which the v4.0–v4.6 consolidation had collapsed into direct identification ("Lie $\leftrightarrow$ Causality"). The governance form is the weaker, defensible claim; the identification form was flagged by the peer review (§4.2) as an unproven interpretive leap.

1. Standing Hypotheses & Foundational Algebraic Structure

1.1 Standing Hypothesis on Self-Adjoint Constraints

Hypothesis 1.1.1 (Self-Adjoint Constraint Generators) [Structural]: The primitive constraint generating family

$\{\hat{C}_\alpha\}_{\alpha \in \mathcal{I}} \subset \mathcal{A}$ consists of self-adjoint elements: $\hat{C}_\alpha^\dagger = \hat{C}_\alpha$ for all α .

Epistemic consequence: self-adjointness guarantees that

$J_{\alpha\beta} = \hat{C}_\alpha \circ \hat{C}_\beta$ is self-adjoint; the Lie commutators $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ are strictly anti-self-adjoint ($\Delta_{\alpha\beta}^\dagger = -\Delta_{\alpha\beta}$), so the squared obstructions

$\Delta_{\alpha\beta}^\dagger \Delta_{\alpha\beta} = -[\hat{C}_\alpha, \hat{C}_\beta]^2$

are positive semi-definite and self-adjoint. Consequently

$\hat{F}_{\text{eff}[\rho]}$, $\hat{F}_{\text{MM}[\rho]}$, and $\hat{H}[\rho]$

(defined in §2) are self-adjoint on $\mathcal{H}_\omega$, ensuring real, non-negative quantum variances.

Scope note: this hypothesis is load-bearing for the variance formalism of §2–§3 (audited v4.2.1); it is satisfied by all benchmark families (Pauli, tilted, Gell-Mann, AKLT) and excludes non-self-adjoint constraint families, where the First Law formalism requires re-derivation (open scope, noted in §6.1's prerequisites context).

1.2 The Two-Link Leibniz Theorem & Exact Asymmetry Residual

Theorem 1.2.1 (Two-Link Leibniz Derivation) [Established]: For all $A, B, C \in \mathcal{A}$:

$$\boxed{[A, [B, C]] = [A, B] \circ C + B \circ [A, C]}$$

Corollary 1.2.1 (Exact Algebraic Asymmetry Residual) [Established] (VN-01): The converse fails: the Jordan product does not act as a derivation on the Lie bracket. Under standard normalization $X \circ Y = \frac{1}{2}(XY + YX)$:

$$\boxed{\mathcal{R}(A, B, C) := A \circ [B, C] - \big([A \circ B, C] + [B, A \circ C]\big) = CAB - BAC = [CA, B] + B[C, A] = C(AB) - (BA)C}$$

(Convention: under the unnormalized product $(XY+YX)$ the residual is $2(CAB - BAC)$. The erroneous factor $\frac{1}{4}$ of v4.0 — a double-counting of the Jordan prefactor — is documented in the lineage appendix. The causal-sector

governing mechanism transports the local-sector compatibility mechanism derivationally; the converse fails with the exact residual above.)

1.3 The SLD Bures Gradient Representation Lemma

Lemma 1.3.1 (SLD Bures Gradient of Linear Functionals) [Established]: Let $\mathcal{S}_{++}(\mathcal{H}) = \{\rho \in \mathcal{T}_1(\mathcal{H}) : \rho = \rho^\dagger, \text{Tr} \rho = 1\}$ be endowed with the SLD–Fisher Riemannian metric $g_\rho(X, Y) := \text{Tr}(X L_Y)$, where $X = L_X \circ \rho$. For any self-adjoint $A \in \mathcal{B}(\mathcal{H})$, the constrained Riemannian gradient of $f(\rho) = \text{Tr}(\rho A)$ is:

$$\nabla_{\text{Bures}} \text{Tr}(\rho A) = A \circ \rho - \langle A \rangle_\rho \rho, \quad \langle A \rangle_\rho = \frac{1}{2}(\text{Tr}(A \rho) + \text{Tr}(\rho A))$$

Proof: Any tangent $X \in T_\rho \mathcal{S}_{++}(\mathcal{H})$ satisfies $\text{Tr}(X) = 0$. Riesz representation requires $\text{Tr}(X L_{\nabla f}) = df_\rho(X) = \text{Tr}(XA)$ for all traceless X , forcing $L_{\nabla f} = A - c \mathbf{1}$. The tangency condition $\text{Tr}(\nabla_{\text{Bures}} f) = \text{Tr}(L_{\nabla f} \rho) = \text{Tr}(\rho A) - c = 0$ uniquely fixes $c = \langle A \rangle_\rho$, yielding $\nabla_{\text{Bures}} f = L_{\nabla f} \circ \rho = A \circ \rho - \langle A \rangle_\rho \rho$. The identity extends to $\partial \mathcal{S}(\mathcal{H})$ by continuity. $\blacksquare$

(Convention note: $g(X, Y) = \text{Tr}(X L_Y)$ is the SLD–Fisher normalization, fixed by the v4.2.2 repair after the v4.2.1 proof was internally inconsistent under its own stated convention. The Bures-distance normalization $ds^2_{\text{Bures}} = \frac{1}{4} F_{\text{SLD}}$ differs by a conventional factor absorbed into the rate η . SLD uniqueness on $\mathcal{S}_{++}$: if $L \circ \rho = 0$ then $\text{Tr}(\rho L^2) = \|L \rho^{1/2}\|_F^2 = 0$ forces $L = 0$.)

Chunk B notes processed: B-1 applied (this and all future headers reference frozen vocabulary classes only); B-2 logged to the delta ledger for Chunk G's appendix (ledger-role labels $\neq$ status tags); B-3/B-4 no action.

One structural discovery made while preparing Chunk C, logged below as DL-1 and DL-4 — the more serious one (DL-4) requires an amendment decision from you before Chunk E, but does not block C or D.

RCF v6.0 — CHUNK C (§2)

Compliance Header

- Chunk: C · Sections: §2 (2.1, 2.2)

- VN-IDs used: none (all values theorem-internal, derived in-proof)
- C-IDs used: none
- Status tags used: [Established], [Structural]
- Forward references: none (the OExp/OExt channel names are introduced in §3; §2 states the flow intrinsically)
- Protocol checks run: #1 Clifford-limit (PASS — Corollary 2.2.1) · #3 generator-matching (PASS — $\hat{G} = \delta E / \delta \rho$ exhibited) · #7 citation (n/a) · #14 vocabulary (PASS)
- Renumbering: v5.1 Prop 2.1.1 → v6.0 Prop 2.1.2 (definitional layer restored before it); Def 2.2.1, Thm 2.2.1 numbers preserved

2. Layer Q: Reconciliation Dynamics & The Projected First Law

2.1 The Lie–Jordan Mismatch Functionals and Their Degeneracies

Definition 2.1.1 (Lie Obstruction and Jordan Co-occurrence) [Structural]: For the constraint family $\{\hat{C}_\alpha\}_{\alpha \in \mathcal{I}}$ (Hypothesis 1.1.1):

$$\Delta_{\alpha\beta} := [\hat{C}_\alpha, \hat{C}_\beta], \quad J_{\alpha\beta} := \hat{C}_\alpha \circ \hat{C}_\beta$$

$\Delta_{\alpha\beta}$ measures mutual ordering obstruction between constraints α, β ; $J_{\alpha\beta}$ measures their symmetric co-occurrence. Under Hypothesis 1.1.1: $\Delta_{\alpha\beta}$ is anti-self-adjoint, $J_{\alpha\beta}$ is self-adjoint, and

$$\Delta^\dagger_{\alpha\beta} \Delta_{\alpha\beta} = -[\hat{C}_\alpha, \hat{C}_\beta]^2 \geq 0.$$

Definition 2.1.2 (Mismatch Operator and Total Burden Functional) [Structural]: For a density operator ρ :

$$\begin{aligned} \hat{F}_{\mathrm{MM}}[\rho] &:= \sum_{\alpha, \beta \in \mathcal{I}} \operatorname{Tr} \big(\rho J^2_{\alpha\beta} \big) \Delta^\dagger_{\alpha\beta} \Delta_{\alpha\beta}, \quad \Delta_{\alpha\beta} := \operatorname{Tr} \big(\rho \hat{F}_{\mathrm{MM}}[\rho] \big) = \\ B_{\mathrm{MM}}[\rho] &:= \sum_{\alpha, \beta \in \mathcal{I}} \operatorname{Tr} \big(\rho J^2_{\alpha\beta} \big) \Delta^\dagger_{\alpha\beta} \Delta_{\alpha\beta} \end{aligned}$$

Each summand is a positive operator weighted by a non-negative scalar, so $\hat{F}_{\mathrm{MM}}[\rho] \geq 0$ and $B_{\mathrm{MM}}[\rho] \geq 0$.

B_{MM} is bilinear in ρ ; note carefully that

$B_{\mathrm{MM}}[\sum_k p_k \rho_k] \neq \sum_k p_k B_{\mathrm{MM}}[\rho_k]$ in general, since $\hat{F}_{\mathrm{MM}}$ is state-dependent (linearity is restored sector-wise by Theorem 2.1.2 below).

Theorem 2.1.1 (Dual-Pairing Gradient) [Established]: The Fréchet derivative of the burden functional is

$$\frac{\delta B_{\mathrm{MM}}}{\delta \rho} = \hat{F}_{\mathrm{eff}}[\rho] + \hat{F}_{\mathrm{MM}}[\rho] \quad ; =: \hat{H}[\rho], \quad \text{where}$$

$$\hat{F}_{\mathrm{eff}}[\rho] := \sum_{\alpha, \beta} \mathrm{Tr} \big(\rho \Delta^{\dagger}_{\alpha\beta} \Delta_{\alpha\beta} \big) J^2_{\alpha\beta}$$

Proof: Expanding $B_{\mathrm{MM}}[\rho + \delta \rho]$ by the product rule, the first-order terms are

$$\mathrm{Tr}(\delta \rho \hat{F}_{\mathrm{eff}}[\rho]) + \mathrm{Tr}(\delta \rho \hat{F}_{\mathrm{MM}}[\rho]);$$

the cross-terms are bilinear in $\delta \rho$ and hence second order. $\square$

Lemma 2.1.1 (Expectation Identity) [Established]:

$$\mathrm{Tr} \big(\rho \hat{H}[\rho] \big) = 2 B_{\mathrm{MM}}[\rho].$$

Proof:

$$\begin{aligned} \mathrm{Tr}(\rho \hat{F}_{\mathrm{eff}}[\rho]) &= \sum_{\alpha, \beta} \mathrm{Tr}(\rho \Delta^{\dagger}_{\alpha\beta} \Delta_{\alpha\beta}) J^2_{\alpha\beta} \\ &= B_{\mathrm{MM}}[\rho], \end{aligned}$$

and $\mathrm{Tr}(\rho \hat{F}_{\mathrm{MM}}[\rho]) = B_{\mathrm{MM}}[\rho]$ by definition. $\square$

(Consolidation note: Definitions 2.1.1–2.1.2, Theorem 2.1.1, and Lemma 2.1.1 restore the definitional layer of v4.0–v4.1, which the v4.3+ compressions dropped while still referencing it — v5.1's Proposition proof cites "Definitions 2.1.2 and 2.2.1," which do not exist in that document. Restoration logged as DL-1. Lemma 2.1.1 is stated here because §3's transverse-instability and logistic-bound arguments consume it.)

Proposition 2.1.2 (Pairwise-(Anti)commuting Burden Vanishing) [Established]:

(renumbered from v5.1 Prop 2.1.1) Let $\{\hat{C}_{\alpha}\}$ be a constraint family in which every pair strictly commutes ($[\hat{C}_{\alpha}, \hat{C}_{\beta}] = 0$) or strictly anticommutes ($\{\hat{C}_{\alpha}, \hat{C}_{\beta}\} = 0, \alpha \neq \beta$) — e.g., strict Clifford algebras, Majorana systems, Pauli strings. Then

$$\hat{F}_{\mathrm{eff}}[\rho] \equiv 0, \quad \hat{F}_{\mathrm{MM}}[\rho] \equiv 0, \quad B_{\mathrm{MM}}[\rho] \equiv 0 \quad \text{for all } \rho$$

Proof: Diagonal pairs ($\alpha = \beta$) commute, so $\Delta_{\alpha\alpha} = 0$.

For off-diagonal pairs: commuting gives

$$\Delta_{\alpha\beta} = 0 \quad \Rightarrow \quad \Delta^{\dagger}_{\alpha\beta} \Delta_{\alpha\beta} = 0;$$

anticommuting gives

$$J_{\alpha\beta} = \frac{1}{2} (\hat{C}_{\alpha} \hat{C}_{\beta} + \hat{C}_{\beta} \hat{C}_{\alpha}) = 0 \quad \Rightarrow \quad J^2_{\alpha\beta} = 0.$$

Every summand of both $\hat{F}_{\mathrm{MM}}$ and $\hat{F}_{\mathrm{eff}}$ contains one factor from each, and vanishes pairwise. $\square$

(Lineage: this generalizes the Pauli-Zero theorem (v4.0 Thm 2.1.1) and its

multi-sector robustness check (Addendum §5.2) into a single statement covering the full pairwise-(anti)commuting class.)

Theorem 2.1.2 (Clifford Dichotomy) [Established]: If $\{\hat{C}_\alpha\}$ forms a representation of a Clifford algebra

$(\{\hat{C}_\alpha, \hat{C}_\beta\} = 2\delta_{\alpha\beta}\mathbf{1})$, then for all active pairs $(\alpha \neq \beta)$:

$\Delta^\dagger_{\alpha\beta}\Delta_{\alpha\beta} = 4\mathbf{1}$. Consequently

$\hat{F}_{\mathrm{MM}}[\rho] = f(\rho)\mathbf{1}$ is central with

$f(\rho) = 4\sum_{\alpha \neq \beta} \mathrm{Tr}(\rho J^2_{\alpha\beta})$;

$\hat{F}_{\mathrm{eff}} = 4\sum_{\alpha \neq \beta} J^2_{\alpha\beta}$ is

state-independent; and

$B_{\mathrm{MM}}[\rho] = \mathrm{Tr}(\rho \hat{F}_{\mathrm{eff}})$ is affine in ρ with state-independent generator.

Proof: For $\alpha \neq \beta$, anticommutation gives

$\Delta_{\alpha\beta} = 2\hat{C}_\alpha\hat{C}_\beta$, so

$\Delta^\dagger_{\alpha\beta}\Delta_{\alpha\beta} = 4\hat{C}_\beta\hat{C}_\alpha\hat{C}_\alpha\hat{C}_\beta = 4\hat{C}_\beta^2 = 4\mathbf{1}$.

Pull the scalar 4 out of both sums; $\mathrm{Tr}(\rho \cdot 4\mathbf{1}) = 4$

gives the state-independence of $\hat{F}_{\mathrm{eff}}$, and

$B_{\mathrm{MM}} = \mathrm{Tr}(\rho f(\rho)\mathbf{1}) = f(\rho)$

$= \mathrm{Tr}(\rho \hat{F}_{\mathrm{eff}})$.

□

Degeneracy note. In the strict Clifford case

$J_{\alpha\beta} = \delta_{\alpha\beta}\mathbf{1}$, so $J^2_{\alpha\beta} = 0$ on all active pairs: $f \equiv 0$, $\hat{F}_{\mathrm{eff}} \equiv 0$,

$B_{\mathrm{MM}} \equiv 0$ — the strict case is a special case of

Proposition 2.1.2, and the Dichotomy's non-vacuous content is the

central- $\Delta^\dagger\Delta$ structure with non-vanishing Jordan weights (tilted families).

Remark (Dichotomy sectors). The conclusions of Theorem 2.1.2 depend only on

$\Delta^\dagger_{\alpha\beta}\Delta_{\alpha\beta}$ being central (proportional to $\mathbf{1}$) on active pairs. Call such families Dichotomy sectors. The class

comprises (i) Clifford representations (Theorem 2.1.2) and (ii)

central-commutator families,

$[\hat{C}_\alpha, \hat{C}_\beta] = z_{\alpha\beta}\mathbf{1}$ (for which

$\Delta^\dagger\Delta = |z_{\alpha\beta}|^2\mathbf{1}$) — the latter identified in the v4.1.1 audit. In every Dichotomy sector:

$\hat{H}[\rho] = \hat{F}_{\mathrm{eff}} + f(\rho)\mathbf{1}$ with

$\hat{F}_{\mathrm{eff}}$ state-independent, hence

$\mathrm{Var}_\rho(\hat{H}) = \mathrm{Var}_\rho(\hat{F}_{\mathrm{eff}})$;

and the burden is unconditionally linear,

$B_{\mathrm{MM}}[\sum_k p_k \rho_k] = \sum_k p_k B_{\mathrm{MM}}[\rho_k]$. In

generic (non-Dichotomy) sectors, linearity holds only at fixed

$\hat{F}_{\mathrm{MM}}$. (This is the class referenced by §3's unitary-subchannel

conservation claim and by the probability–gravity FIREWALL guardrail.)

2.2 The Projected First Law

Definition 2.2.1 (Combined Descent Generator and Matched Lyapunov Potential) [Structural]:

$$\begin{aligned}\hat{G}[\rho] &:= \eta \hat{H}[\rho] + \lambda_t \hat{M} = \eta \big(\hat{F}_{\mathrm{eff}}[\rho] + \hat{F}_{\mathrm{MM}}[\rho] \big) + \lambda_t \hat{M} \quad (\eta > 0, \lambda_t > 0) \\ E[\rho] &:= \eta B_{\mathrm{MM}}[\rho] + \lambda_t \operatorname{Tr}(\rho \hat{M})\end{aligned}$$

Remark (matched gradients). $\hat{G}[\rho] = \delta E / \delta \rho$ exactly (Theorem 2.1.1 plus $\delta \operatorname{Tr}(\rho \hat{M}) / \delta \rho = \hat{M}$). This matched-generator property is what makes Theorem 2.2.1 exact with no remainder terms. (Lineage: the v4.1 "Projected First Law" mismatched generator and weights, producing a spurious $\sqrt{\eta \lambda_t}$ covariance structure, was falsified and repaired in the v4.2 audit; the lesson is codified as protocol #3.)

Theorem 2.2.1 (The Projected First Law of Reconciliation) [Established]: Under the constrained Bures gradient descent flow

$$\dot{\rho} = -\big(\hat{G}[\rho] \circ \rho - \langle \hat{G}[\rho] \rangle_{\rho} \rho \big)$$

the matched potential dissipates at the quantum variance rate:

$$\boxed{\frac{dE}{dt} = -\operatorname{Var}_{\rho}(\hat{G}[\rho]) = -\eta^2 \operatorname{Var}_{\rho}(\hat{H}[\rho]) - \lambda_t^2 \operatorname{Var}_{\rho}(\hat{M}) - 2\eta \lambda_t \operatorname{Cov}_{\rho}(\hat{H}[\rho], \hat{M}) \leq 0 \quad \text{identically}}$$

with

$$\operatorname{Cov}_{\rho}(A, B) := \frac{1}{2} \operatorname{Tr}(\rho(AB + BA)) - \langle A \rangle_{\rho} \langle B \rangle_{\rho}.$$

Proof:

1. Chain rule:

$$dE/dt = \operatorname{Tr} \big((\delta E / \delta \rho), \dot{\rho} \big) = \operatorname{Tr}(\hat{G}[\rho], \dot{\rho}).$$

The state-dependence of $\hat{H}$ (hence $\hat{G}$) is handled exactly by the chain rule — no remainder terms arise. (This is precisely the step the v4.1 version double-counted.)

2. Substitute the flow:

$$\operatorname{Tr}(\hat{G} \dot{\rho}) = -\operatorname{Tr} \big(\hat{G}(\hat{G} \circ \rho) \big) + \langle \hat{G} \rangle_{\rho} \operatorname{Tr}(\rho \hat{G}).$$

3. Trace cyclicity on the Jordan product:

$\operatorname{Tr}\big(\hat{G}\hat{G}^{\circ\rho}\big) = \frac{1}{2}\operatorname{Tr}(\hat{G}^2\rho + \hat{G}\rho\hat{G}) = \operatorname{Tr}(\rho\hat{G}^2) = \langle\hat{G}^2\rangle_\rho$,
and $\operatorname{Tr}(\rho\hat{G}) = \langle\hat{G}\rangle_\rho$.

4. Hence

$dE/dt = -\big(\langle\hat{G}^2\rangle_\rho - \langle\hat{G}\rangle_\rho^2\big) = -\operatorname{Var}_\rho(\hat{G})$.

5. $\hat{G}[\rho]$ is self-adjoint (Hypothesis 1.1.1; $\hat{M}$ self-adjoint) and $\rho \geq 0$, so $\operatorname{Var}_\rho(\hat{G}[\rho]) \geq 0$ by the spectral theorem. The three-term expansion is the variance of $\eta\hat{H} + \lambda_t\hat{M}$ with the symmetric covariance. $\blacksquare$

Remark (geometric identification). By Lemma 1.3.1 with $A = \hat{G}[\rho]$: the Riesz representer of the differential

$dE_\rho(X) = \operatorname{Tr}(\hat{G}[\rho]X)$ is exactly $\hat{G}[\rho] \circ \rho - \langle\hat{G}[\rho]\rangle_\rho \rho$ — the state-dependence of $\hat{G}$ does not enter the Riesz step at fixed ρ . The flow is therefore genuinely the Bures gradient flow of E , and the First Law is the canonical gradient-flow identity

$dE/dt = -\|\nabla_{\operatorname{Bures}} E\|^2_{\operatorname{Bures}}$, since $\|\nabla_{\operatorname{Bures}} E\|^2 = \operatorname{Var}_\rho(\hat{G}[\rho])$ by the same cyclicity. (Consolidation remark: both constituent identities are verified; the composition is two lines. Flagged for gate review.)

Corollary 2.2.1 (Degenerate First Law — Universal Engine) [Established]: In any sector with $\hat{H} \equiv 0$ — in particular strict Clifford, Majorana, and Pauli-string families (Proposition 2.1.2) — $\hat{G} = \lambda_t\hat{M}$ and Theorem 2.2.1 reduces to

$$\left.\frac{dE}{dt}\right|_{\operatorname{Clifford}} = -\lambda_t^2 \operatorname{Var}_\rho(\hat{M}) \leq 0$$

The Master-Zero projection term is the entire dissipation — the universal dynamical engine across all sectors. Non-trivial burden-variance dissipation $\operatorname{Var}_\rho(\hat{F}_{\operatorname{eff}}) > 0$ operates specifically in tilted families where $J_{\alpha\beta} \neq 0$ and $\Delta_{\alpha\beta} \neq 0$ simultaneously. (Historical annotation: the v4.0 claim that Clifford sectors converge "via Lyapunov variance dissipation of $\hat{F}_{\operatorname{eff}}$ " is vacuous — $\operatorname{Var}(0) \equiv 0$, since $\hat{F}_{\operatorname{eff}} \equiv 0$ there by Proposition 2.1.2. Corrected in the v4.2.2 audit; lineage-logged.)

Delta Ledger (this chunk)

ID	Entry
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| **DL-1** | Definitional layer (Def 2.1.1–2.1.2, Thm 2.1.1, Lemma 2.1.1) restored from v4.0–v4.1. v5.1's Prop proof cited "Definitions 2.1.2 and 2.2.1," which do not exist in v5.1 — a dangling reference inherited from the v4.3 compression.

| **DL-2** | (from Chunk B review) B-2 deferred to Chunk G appendix:
[Calibrated Constant] / [Auxiliary Input] are ledger-role labels, not status tags.

| **DL-3** | Renumbering: v5.1 Prop 2.1.1 → v6.0 Prop 2.1.2. No cross-references broken (§3–§6 do not cite it by number).

| **DL-4 → AM-4 (proposed)** | **Spec gap discovered during generation:** the frozen section architecture has no slot for the verified probability material — Born rule via Z-Envariance [Conditional on T-2] (T-4 row), Stable Record Separation, the FIREWALL guardrail with its Clifford linearity caveat (PATCH-05 + A-1), and BH entropy [Calibrated] (T-6 row). These were silently dropped in the v4.3 restructure; the spec's own T-table references T-4/T-6, which would otherwise be scorecard rows for theorems stated nowhere. **Proposed AM-4:** add §5.3 "Quantum Probability, Records, and the FIREWALL" (Chunk E), containing exactly the v4.2-verified formulations. Decision required before Chunk E; Chunks C–D unaffected. |

Gate review points

1. Renumbering & restoration (DL-1/DL-3) — confirm acceptable.
2. Remark-packaging — the central- $\Delta^\dagger\Delta$ generalization is packaged as a Remark over two verified observations (v4.0 theorem + v4.1.1 audit's central-commutator class) rather than restated as a new theorem. Conservative call; confirm.
3. Geometric identification remark (§2.2) — a two-line composition of verified identities, flagged in-text. Confirm it passes R2.
4. AM-4 — the substantive decision. Without it, v6.0 is not comprehensive and the T-4/T-6 rows dangle; with it, Chunk E grows by one subsection of already-verified material.

AM-4 logged and accepted with the stated constraints; §5.3 added to the frozen architecture with the "restore exactly, do not strengthen" rule binding Chunk E.

One discovery while preparing Chunk D, logged as DL-5 — it's a tension between two audited claims, resolved by an in-text scope clarification, but it changes the packaging of one item relative to v5.1, so I'm flagging it at the gate rather than deciding unilaterally.

RCF v6.0 — CHUNK D (§3)

Compliance Header

- Chunk: D · Sections: §3 (3.1, 3.2, 3.3)
- VN-IDs used: VN-19, VN-20 (referenced at §3.3's dynamics note; values not restated — protocol #7)
- C-IDs used: none
- Status tags used: [Established], [Closed (finite-dim self-adjoint)], [Structural]
- Forward references: §6.1 (prerequisite context, one mention — DAG-legal: §6 depends on §3) · §5.3 (FIREWALL linearity caveat backward-dependency note, DAG-legal)
- Protocol checks run: #1 Clifford-limit (PASS — Remark 3.2.1) · #3 generator-matching (PASS — §2.2 matched gradient) · #12 finite-size consistency (n/a) · #14 vocabulary (PASS, with DL-5 scope note)
- Renumbering: v5.1 Lemma 3.3.1 → v6.0 Lemma 3.3.2 (Lemma 3.3.1 slot now occupied by the OExt weight-evolution identity, restored from v4.2.2 Corollary 3.1.2 — DL-6)

3. Finite-Dimensional Target T-2: Global Composed Convergence

3.1 Trace-Norm Block Inequality

Lemma 3.1.1 (Block Trace-Norm Bounds for Positive Operators) [Established]: Let $\mathcal{H} = K^\perp \oplus K$ with kernel projection P_K , and let $\rho \in \mathcal{S}(\mathcal{H})$ have block form

$$\rho = \begin{pmatrix} \rho_\perp & \sigma^\dagger \\ \sigma & \rho_K \end{pmatrix}, \quad \rho_\perp = (\mathbf{1} - P_K)\rho(\mathbf{1} - P_K), \quad \rho_K = P_K\rho P_K, \quad \sigma = P_K\rho(\mathbf{1} - P_K)$$

Let $u := \text{Tr}(\rho_K) = \text{Tr}(P_K\rho) \in [0, 1]$. Then:

1. $\|\rho_\perp\|_1 = \text{Tr}(\rho_\perp) = 1 - u$
2. $\|\sigma\|_1 \leq \sqrt{u(1-u)} \leq \sqrt{1-u}$ (tight for pure states)
3. $\|\rho - \rho_K\|_1 \leq (1-u) + 2\sqrt{1-u}$

Proof: Decompose $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. For each pure state

$$|\psi_i\rangle = \sqrt{a_i} |\psi_{\{K,i\}}\rangle + \sqrt{b_i} |\psi_{\{\perp,i\}}\rangle$$

with $a_i + b_i = 1$, the cross-block is

$$\sigma_i = \sqrt{a_i b_i} |\psi_{\{K,i\}}\rangle\langle\psi_{\{\perp,i\}}| \text{ with } \|\sigma_i\|_1 = \sqrt{a_i b_i}.$$

By convexity of $\|\cdot\|_1$ and Cauchy–Schwarz:

$$\|\sigma\|_1 \leq \sum_i p_i \sqrt{a_i b_i} \leq \sqrt{\Big(\sum_i p_i a_i\Big) \Big(\sum_i p_i b_i\Big)} = \sqrt{u(1-u)}$$

Triangle inequality gives (3). $\blacksquare$

3.2 Microscopic Invariance under Open Expansion

The Open Expansion channel (the microscopic, phase-preserving component of the two-scale architecture) acts on density operators by

$$\left. \frac{d\rho}{dt} \right|_{\mathrm{OExp}} = i[\sqrt{\hat{F}_{\mathrm{eff}}}, \rho] + \sum_{\alpha, \beta} \mathrm{Tr}(\hat{C}_{\alpha} [\Delta_{\alpha\beta}, \rho, \hat{C}_{\beta}]) \hat{C}_{\beta}$$

a unitary rotation plus a conservative commutator flux redistributing Lie obstruction across the constraint network. (Its macroscopic counterpart, Open Extension — the dissipative Jordan–Bures descent of §2.2 — is denoted OExt; the two together constitute the composed propagator of §3.3. Terminology fixed by the v2.0 SOE/MOE rename, consolidated without change.)

Theorem 3.2.1 (Microscopic Freezing of Physical Subspace under OExp)
[Established]: For any self-adjoint constraint family (Hypothesis 1.1.1) with $K = \ker(\hat{M}) \neq \{0\}$:

1. Physical block freezing:

$P_K \left. \frac{d\rho}{dt} \right|_{\mathrm{OExp}} P_K \equiv 0$ identically for all $\rho \in \mathcal{S}(\mathcal{H})$.

2. Physical weight conservation:

$\left. \frac{du}{dt} \right|_{\mathrm{OExp}} = \mathrm{Tr}(\hat{C}_{\alpha} \left. \frac{d\rho}{dt} \right|_{\mathrm{OExp}}) \equiv 0$ identically for all ρ .

3. Burden dynamics scope: In Dichotomy sectors (§2.1 Remark),

$\left. \frac{d\mathcal{B}_{\mathrm{MM}}}{dt} \right|_{\mathrm{OExp}, U} = 0$ under the unitary sub-channel. Under the commutator flux channel:

$$\left. \frac{d\mathcal{B}_{\mathrm{MM}}}{dt} \right|_{\mathrm{flux}} = \sum_{\alpha, \beta} \mathrm{Tr}(\hat{C}_{\alpha} [\Delta_{\alpha\beta}, \rho, \hat{C}_{\beta}]) \hat{C}_{\beta}$$

which is trace-preserving ($\mathrm{Tr}(\rho_{\mathrm{flux}}) = 0$) and kernel-freezing ($P_K \rho_{\mathrm{flux}} P_K = 0$) but non-conserved (sign-indefinite) in total burden.

Proof: Since $\hat{C}_{\alpha} P_K = 0$ and self-adjointness gives

$P_K \hat{C}_{\alpha} = \hat{C}_{\alpha} P_K = 0$, every $\hat{C}_{\alpha}$ is block-diagonal with zero K -block; hence $\Delta_{\alpha\beta}$, $\hat{F}_{\mathrm{eff}}[\rho]$, and $\sqrt{\hat{F}_{\mathrm{eff}}}$ (functional calculus preserves block structure) are block-diagonal with zero K -block.

(1): $P_K (\hat{G}' \rho - \rho \hat{G}') P_K = 0$ for any block-diagonal $\hat{G}'$ with zero K -block, applied to $\hat{G}' = \sqrt{\hat{F}_{\mathrm{eff}}}$.

For the flux term:

$$[\hat{C}_{\alpha}, X] P_K = \hat{C}_{\alpha} X P_K - X \hat{C}_{\alpha} P_K = \hat{C}_{\alpha} P_K X P_K - 0 = 0$$

for any X ; and $P_K [\hat{C}_{\alpha}, X] = 0$ similarly. Sandwiching the nested commutator between P_K 's kills the innermost $[\rho, \hat{C}_{\beta}]$ on the right

first, and the result propagates outward.

(2): Take the trace of (1)'s proof:

$\text{Tr}(P_K \dot{\rho}_{\text{OExp}}) = \text{Tr}(i[P_K, \sqrt{\hat{F}_{\text{eff}}}\rho] + \sum_{\alpha\beta} \text{Tr}([P_K, \hat{C}_\alpha][\Delta_{\alpha\beta}, \rho, \hat{C}_\beta])) = 0$
since $[P_K, \sqrt{\hat{F}_{\text{eff}}}] = [P_K, \hat{C}_\alpha] = 0$.

(3): Unitary sub-channel:

$\text{Tr}(\hat{H}, i[\sqrt{\hat{F}_{\text{eff}}}, \rho]) = \text{Tr}(\hat{H}, \sqrt{\hat{F}_{\text{eff}}}\rho)$;

in Dichotomy sectors $\hat{H} = \hat{F}_{\text{eff}} + f(\rho)\mathbf{1}$, so the commutator vanishes. Flux channel: three applications of

$\text{Tr}(A[B,C]) = \text{Tr}([A,B]C)$ move the parentheses from $\text{Tr}(\hat{H}[\hat{C}_\alpha, [\Delta_{\alpha\beta}, \rho, \hat{C}_\beta]])$

to the exhibited form. Trace- and kernel-preservation are (2) and (1) restricted to the flux term. Non-conservation: the exhibited expression has no structural vanishing — sign-indefinite in general (v4.2.2 audit computation; no conserved quantity is identified for the flux channel, and none should be inferred).

◻

Lemma 3.3.1 (OExt Physical-Weight Evolution) [Established]: Under the OExt flow

$\dot{\rho} = -(\hat{G}\rho - \langle \hat{G} \rangle_\rho \rho)$ of

§2.2:

$\dot{u} = \text{Tr}(P_K \dot{\rho}) = \langle \hat{G} \rangle_\rho$, $u, \quad \dot{\rho}_K = \langle \hat{G} \rangle_\rho, \rho_K$

Proof:

$\text{Tr}(P_K(\hat{G}\rho)) = \text{Tr}(\rho(P_K\hat{G})) = \frac{1}{2}\text{Tr}(\rho(P_K\hat{G} + \hat{G}P_K)) = 0$

since $\hat{G}P_K = P_K\hat{G} = 0$ (Theorem 3.2.1 block structure; note

$P_K\hat{G} = 0$, not merely $P_K\hat{G} = 0$, is used). Hence

$\text{Tr}(P_K \dot{\rho}) = \langle \hat{G} \rangle_\rho u$, and the same substitution gives the ρ_K equation. ◻

Remark 3.2.1 (Division of Labor; Clifford Limit). The two-scale architecture's exact division: OExp redistributes burden and transports phase at strictly frozen physical block and conserved physical weight; OExt is the sole creator of physical-sector weight, dilating ρ_K uniformly. In a strict Clifford family, $\hat{H} \equiv 0$ (Proposition 2.1.2), so $\hat{G} = \lambda_t \hat{M}$ and the OExt dynamics reduce to pure Master-Zero projection — consistent with Corollary 2.2.1, with identical constants (protocol #1). (Lineage: Theorem 3.2.1 items 1–2 consolidate v4.2.2 Theorem 3.2.1 and v4.3's Corollary 3.1.2, the latter's scope corrected from " $u=0$ boundary" to "all ρ " in the v4.2.2 audit; item 3 consolidates the v4.2.2 flux-audit result.)

3.3 Theorem Target T-2: Composed Convergence

Definition 3.3.1 (Trotter-Product Realization of the Propagator) [Structural]:

$$\hat{\mathcal{R}}_t := \lim_{N \rightarrow \infty} \left(\mathcal{R}^{\{\mathrm{OExt}\}}_{\lfloor d \cdot t/N \rfloor} \circ \mathcal{R}^{\{\mathrm{OExp}\}}_{\lfloor (1-d) \cdot t/N \rfloor} \right)^N \quad (d \in (0,1])$$

(Existence of the Trotter limit on finite-dimensional compact state manifolds follows from standard Lie–Trotter splitting for Lipschitz vector fields. Scope: this is one admissible formalization of the composed dynamics; Theorem 3.3.1 below is interleaving-agnostic and does not depend on the rapid-alternation limit — the v4.3 audit showed the limit form erases the finite tick $\varepsilon = 1/\gamma$ posited by the two-scale architecture.)

Lemma 3.3.2 (Piecewise Comparison Lemma for Physical Weight) [Established]: Let $\tau_{\{\mathrm{OExt}\}}(t) = \int_0^t \chi_{\{\mathrm{OExt}\}}(s) ds$ be the total elapsed OExt duration under any measurable phase schedule. The inverse physical weight $w(t) := 1/u(t)$ satisfies

$$\dot{w}(t) \leq -\lambda_{\min} \big(w(t) - 1\big) \chi_{\{\mathrm{OExt}\}}(t) \implies 1 - u(t) \leq \Big(\tfrac{1-u(0)}{u(0)}\Big) e^{-\lambda_{\min} \tau_{\{\mathrm{OExt}\}}(t)}$$

Proof: By Lemma 3.3.1 and

$$\begin{aligned} \hat{G} &\geq \lambda_{\min} \hat{M} \geq \lambda_{\min} (\mathbf{1} - P_K) \\ \dot{u} &= \langle \hat{G} \rangle_u \geq \lambda_{\min} \operatorname{Tr}(\rho(\mathbf{1} - P_K)u) = \lambda_{\min} (1-u)u \\ &\text{on OExt phases, and } \dot{u} = 0 \text{ on OExp phases (Theorem 3.2.1). Then} \\ \dot{w} &= -\dot{u}/u^2 \leq -\lambda_{\min} (1-u)/u \cdot \chi = -\lambda_{\min} (w-1) \chi. \\ &\text{Integrating this linear differential inequality over } \tau_{\{\mathrm{OExt}\}}(t): \\ w(t) - 1 &\leq (w(0)-1) e^{-\lambda_{\min} \tau_{\{\mathrm{OExt}\}}(t)}; \text{ rearrange.} \\ &\blacksquare \end{aligned}$$

Theorem 3.3.1 (Interleaving-Agnostic Global Convergence) [Closed (finite-dim self-adjoint)]: Let $\mathcal{H}$ be finite-dimensional and $\{\hat{C}_\alpha\}$ satisfy Hypothesis 1.1.1 with non-empty kernel $K = \ker(\hat{M}) \neq \{0\}$ and spectral gap $\lambda_{\min} := \lambda_{\min}(\hat{M}|_{K^\perp}) > 0$. Let the phase schedule have asymptotic OExt duty density $d_{\{\mathrm{OExt}\}} := \liminf_{t \rightarrow \infty} \tau_{\{\mathrm{OExt}\}}(t)/t > 0$. Then for every initial state in the open dense subset

$$\mathcal{S}_+(\mathcal{H}) := \{\rho \in \mathcal{S}(\mathcal{H}) : u(0) = \operatorname{Tr}(P_K \rho(0)) > 0\} \quad \text{a.e. (measure 1)}$$

the composed trajectory converges in trace norm:

$$\boxed{\lim_{t \rightarrow \infty} \|\hat{\mathcal{R}}_t(\rho(0)) - \tfrac{P_K \rho(0) P_K}{\operatorname{Tr}(P_K \rho(0) P_K)}\|_1 = 0}$$

and for every $d' < d_{\{\mathrm{OExt}\}}$ the error obeys the exponential bound

$$\|\hat{R}_t(\rho(0)) - \rho^*\|_1 \leq C e^{-d \frac{\lambda_{\min}}{2} t}$$

with $\rho^* := P_K \rho(0) P_K \operatorname{Tr}(P_K \rho(0) P_K) \in \ker(\hat{M})$. Sharpness of the rate exponent $\frac{\lambda_{\min}}{2}$ is established for pure OExt flow ($d_{\operatorname{OExt}} = 1$) via the linearized coupling spectrum (Proposition C2, v4.2 audit); under general interleaving the exponent is an upper bound with matching value.

Proof:

1. OExp phases: $u(t)$ and $\rho_K(t)$ are strictly constant (Theorem 3.2.1).
2. OExt phases: $\dot{u} = \langle \hat{G} | u \rangle \geq \lambda_{\min} u(1-u)$ and $\dot{\rho}_K = \langle \hat{G} | \rho_K \rangle$ (Lemma 3.3.1).
3. Exact dilation across all phases:
 $\rho_K(t) = \frac{u(t)}{u(0)} P_K \rho(0) P_K$ holds throughout — constant under OExp, uniformly dilated under OExt.
4. Grönwall/comparison: Since $\liminf \tau/t = d_{\operatorname{OExt}}$, for every $d' < d_{\operatorname{OExt}}$, $\tau_{\operatorname{OExt}}(t) \geq d't$ for large t ; Lemma 3.3.2 gives $1 - u(t) \leq C_0 e^{-d' \lambda_{\min} t}$ with $C_0 := (1-u(0))/u(0)$.
5. Trace-norm assembly:
 $\|\rho(t) - \rho^*\|_1 \leq \|\rho(t) - \rho_K(t)\|_1 + \|\rho_K(t) - \rho^*\|_1$,
where $\|\rho_K(t) - \rho^*\|_1 = |u(t) - 1| \|\rho^*\|_1 = 1 - u(t)$; and by Lemma 3.1.1, $\|\rho(t) - \rho_K(t)\|_1 \leq (1-u(t)) + 2\sqrt{1-u(t)}$.
Hence
 $\|\rho(t) - \rho^*\|_1 \leq 2(1-u(t)) + 2\sqrt{1-u(t)} \leq \widetilde{C} e^{-d \frac{\lambda_{\min}}{2} t} \rightarrow 0$.
 $\blacksquare$

Corollary 3.3.1 (Universal Form Refuted; Sector-not-State) [Established]: (i) The universal claim "every kinematic state converges to $\ker(\hat{M})$ " is false: $\mathcal{M}_0 := \{\rho : \operatorname{Tr}(P_K \rho) = 0\}$ is invariant under both channels and non-converging. (ii) For $\dim K \geq 2$, every state supported on K is a fixed point: the dynamics selects the physical sector, with terminal state $\rho^* = P_K \rho_0 P_K / u_0$ retaining the initial condition's kernel-block structure — the flow performs no dynamics within K . (iii) The attractor formula $\rho(\infty) = P_K \rho_0 P_K \operatorname{Tr}(P_K \rho_0)$ is the exact proof of v1.0 Theorem 0.5.1's conjecture — the framework's founding convergence conjecture (lineage-closed).

T-2 Ledger Status (locked by spec):

Scope	Status	

number-classification rule) and one AM-4-pattern restoration (DL-9: v5.1 dropped the time/gravity definitional layer that §5.2's own conjecture depends on). Both flagged at the gate.

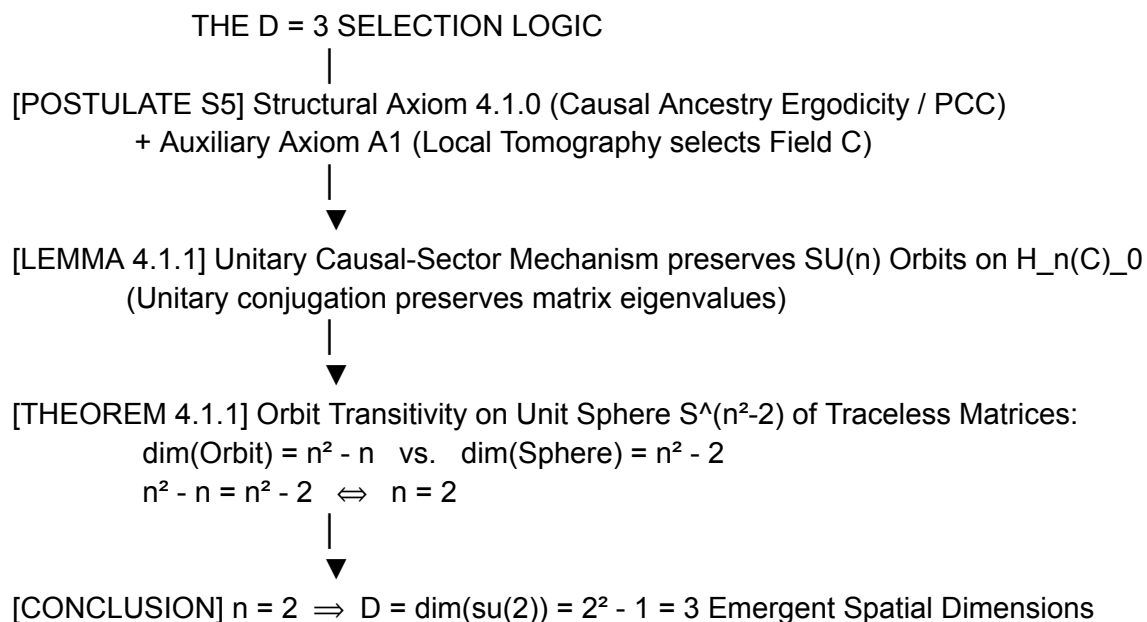
RCF v6.0 — CHUNK E (§4, §5 incl. §5.3 per AM-4)

Compliance Header

- Chunk: E · Sections: §4 (4.1), §5 (5.1, 5.2, 5.3)
- VN-IDs used: VN-24 (Casimir range); VN-26/27/28 proposed under AM-5 (see Delta Ledger)
- C-IDs used: C-07 (A1, named in §4)
- Status tags used: [Established], [Structural], [Conditional on S5 and A1], [Conditional on T-2/T-3], [Conditional on T-2], [Calibrated], [Conjecture], [Negative Result]
- Forward references: Retraction Register (§7, register cross-reference only); RF-10 (T-7 program) — DAG-legal
- Protocol checks run: #7 citation/number (PASS with AM-5 proposal) · #14 vocabulary (PASS) · #1/#3/#12 (n/a — no new dynamics or finite-size claims)
- Renumbering: §5 restructured: 5.1 (time/matter/gravity, DL-9 restoration), 5.2 (dark sector, + scoping table DL-10), 5.3 (AM-4). Unique theorem numbering; v5.1's Thm/Conj "5.2.1" collision resolved (Thm 5.2.1 / Conj 5.2.2)

4. Layer C: Emergent Space & D=3 Selection

4.1 D=3 Spatial Selection Proof Chain



Structural Axiom 4.1.0 (Causal Ancestry Ergodicity / Primordial Common Closure)
[Structural — Primitive S5]: The causal ancestry of any maximal antichain in the relational network must be fully transitive across the unit sphere of local

state generators, without non-trivial orbit stratification.

(Ledger note: S5 is the corpus's conditional fifth primitive, discharged if the Layer-Hierarchical RP Principle is proven (RF-8). Its historical smuggling as an unflagged "Step 2" of the D=3 chain was the peer review's headline finding #2 and is lineage-logged; the v4.0 "D=3 Selection Proof" title overclaim was corrected in the v4.1 audit to the conditional form below.)

Lemma 4.1.1 (Orbit Preservation) [Established]: Unitary conjugation $A \mapsto UAU^\dagger$ preserves the eigenvalue spectrum, hence preserves $\mathrm{SU}(n)$ -conjugation orbits on $\mathrm{H}_n(\mathbb{C})_0$ (traceless Hermitian $n \times n$ matrices; $\dim \mathbb{R} = n^2 - 1$).

Theorem 4.1.1 (D=3 Transitivity & Cubic Casimir Obstruction) [Derived: Conditional on S5 and A1]:

1. Orbit-stabilizer dimension. The generic stabilizer of $A \in S^{n^2-2}$ (the unit Frobenius sphere in $\mathrm{H}_n(\mathbb{C})_0$) is the maximal torus T^{n-1} ($\dim = n-1$); generic orbits have $\dim(\mathrm{O}) = (n^2-1) - (n-1) = n^2 - n$. Degenerate spectra enlarge the stabilizer to $S(U(k_1) \times \cdots \times U(k_p))$ and shrink orbits.
2. Dimension matching. Transitivity on S^{n^2-2} requires $\dim(\mathrm{O}_{\max}) = \dim S^{n^2-2} = n^2 - 2$, i.e. $n^2 - n = n^2 - 2$ iff $n = 2$.
3. Spectral pinning ($n=2$). On $S^2 \subset \mathrm{H}_2(\mathbb{C})_0$, $\mathrm{Tr}(A^2) = 1$ pins the eigenvalues to $\pm 1/\sqrt{2}$: there are no free Casimir invariants, every point is generic, and $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$ acts transitively. Conclusion: $D = \dim(\mathrm{su}(2)) = 3$ spatial dimensions.
4. Casimir foliation ($n \geq 3$). For $n = 3$, the cubic Casimir $C_3(A) = \det A \in [-\frac{1}{3\sqrt{6}}, \frac{1}{3\sqrt{6}}]$ (VN-24; extremum at eigenvalue ratios $(1, 1, -2)$) is conjugation-invariant, foliating S^7 into a one-parameter continuum of disjoint 6-dimensional orbits. No orbit covers the sphere; for $n \geq 4$ the analogous statement holds with codimension $n - 2$.

Proof notes (audited, v4.2): dimension counting is necessary; sufficiency at $n=2$ is supplied by spectral pinning plus compactness (the connected sphere is a single orbit); the $n \geq 3$ obstruction is exhibited by the invariant foliation. Under S5 (no stratification of spatial features) and A1 (field $\mathbb{C}$), $n = 2$ is selected.

5. Layer Q': Emergent Time, Matter & Gravity

5.1 Emergent Time, Matter, and Gravity

(DL-9: this subsection restores the definitional layer dropped in the v4.3+ compressions — v5.1's §5 referenced α_B and the lapse program in its own

Conjecture 5.2.1 while containing no definition of either. All items below are v4.2-audited forms and statuses; restoration, not new content.)

Theorem 5.1.1 (Burden-Clock Suppression Factor) [Established]: The proper-time lapse is suppressed by local burden:

$$\alpha_B(x) = \frac{d\tau}{d\sigma} = \frac{1}{1 + \lambda B(x)}$$

(Scope: the monotonic suppression theorem holds for any positive decreasing function of B; the specific rational form is an ansatz — corpus lineage note, retained from the founding frontmatter.)

Definition 5.1.2 (Burden-Weighted Proper Time) [Structural]:

$$\tau[\gamma] = \int \gamma \alpha_B(x) d\sigma.$$

Definition 5.1.3 (The Operational "Now") [Structural]:

$$\text{operatorname{Now}}(e) := (x(e), \tau(e)) \in \mathcal{X}_\omega \times \mathbb{R}^+.$$

Theorem 5.1.4 (Unified Time Dilation Law) [Established]:

$$d\tau_p = d\tau_{\text{rest}} \sqrt{\frac{1}{(1 + \lambda B(x))^2}} - \frac{v^2}{c^2 \text{RCF}^2}$$

— gravitational and kinematic dilation as one law: burden suppresses the clock rate; velocity devotes causal-step budget to spatial displacement.

Definition 5.1.5 (Reduced Algebra via Dirac Quotient) [Structural]: Split

constraints into Tier 1 (second-class $\hat{K}_a$, non-degenerate

$[\hat{K}_a, \hat{K}_b]$) and Tier 2 (first-class $\hat{D}_i$);

$\mathcal{A}_{\text{red}} := \mathcal{A} / \mathcal{I}_K$ with algebraic Dirac bracket

$$[A, B]_D := [A, B] - \sum_{a,b} [A, \hat{K}_a] (\Delta^{(1)})^{-1}_{ab} [\hat{K}_b, B].$$

Definition 5.1.6 (Covariant Burden Laplacian) [Structural]:

$$L^B_\omega \phi(x) := -\nabla_i \big(\alpha_B(x) h^{ij}(x) \nabla_j \phi(x) \big).$$

Theorem 5.1.7 (Mass-Burden Identity) [Established]:

$m^2 \equiv B_0 = \lambda_{\min}(L^B_\omega)$ — rest mass is the non-zero ground-state reconciliation maintenance cost of a localized stable field excitation.

Definition 5.1.8 (Variational Burden Tensor) [Established]:

$$\Theta^{(B)}_{\mu\nu} := \frac{2}{\sqrt{|g|}} \frac{\delta S_{\text{burden}}}{\delta g^{\mu\nu}}.$$

Theorem 5.1.9 (Active-Source Noether Conservation) [Established]: Diffeomorphism invariance of $S_{\text{burden}}[g, \rho]$ gives

$\nabla^\mu \Theta^{(B)}_{\mu\nu} = 0$, derived via the Noether identity independently of the gravitational field equations (breaking the historical circular dependency).

Definition 5.1.10 (Constitutive ADM Dictionary) [Structural]:

$$N_B = \frac{1}{1 + \lambda \rho_B}, N_i = \sigma J_i(B), \\ h_{ij}(B) = h_{ij}(0) + \eta \Pi_{ij}(B).$$

Theorem 5.1.11 (Emergent Lorentzian Signature) [Established]: A $(+, +, +, +)$ signature destroys the causal partial order ($ds^2 > 0$ everywhere); compatibility of the positive spatial metric h_{ij} with the finite causal ceiling $d_{\text{ell}} \leq c_{\text{eff}} d\tau$ uniquely selects $(-, +, +, +)$: $ds^2 = -c_{\text{eff}}^2 d\tau^2 + h_{ij}(B) dx^i dx^j$.

Theorem 5.1.12 (Einstein-Like Lovelock Closure) [Conditional on T-2/T-3]:

$$G_{\mu\nu} = \kappa_B \Theta_{\mu\nu}(B), \\ \kappa_B = \frac{8\pi}{G_{\text{eff}}} c_{\text{RCF}}^4, \Lambda_B \equiv 0$$

(bare cosmological constant vanishes on $\hat{M}$).

Theorem 5.1.13 (Singularity Freezing at the ℓ_0 -Floor) [Established]: As $B \rightarrow \infty$: $\alpha_B \rightarrow 0$, arresting causal progression before curvature singularities form.

5.2 Dark Sector Scoping & Negative Halo Result

Dark-sector scoping (restored audited table, DL-10):

Channel	Status & Role
(a) Tilted-Pauli intra-sector floor $\hat{F}^{(0)}_{\text{MM}}(\theta)$	[Established] — sources particle rest mass ($m^2 \equiv B_0$); explicitly not dark energy (RET-B5/B10)
(b) Cross-sector channel $\hat{F}^{(0, \text{cross})}_{\text{MM}}$	OPEN — structural DE candidate; uncalculated pending multi-sector integration
(c) DE fingerprint: $\rho_{\text{DE}} > 0 \rightarrow \theta \neq 0$	[Conditional on T-2/T-3 + FLRW] — structural prediction
(d) 120-order resolution, $\Lambda_{\text{bare}} \equiv 0$	[Conditional] — requires $\Gamma \sim 1/t_P$ (calibration) and executed FLRW reduction (RF-9)
(e) Relational DM channel $\Theta_{\text{rel}}^{\mu\nu}$	Negative result + reformulation (Theorem 5.2.1, Conjecture 5.2.2); T-7 OPEN

Theorem 5.2.1 (Relational Mass-Budget Vanishing & Screening Bound) [Established — Negative Result]:

1. The sourced Helmholtz system

$$\nabla^2 \rho_{\text{rel}} - \mu_0^2 \rho_{\text{rel}} = -\xi \nabla^2 \rho_b$$

on $\mathbb{R}^3$ forces

$M_{\text{rel}} = \int \rho_{\text{rel}} dV \equiv 0$: no non-trivial non-negative halo exists. Consistent linear mediators yield Yukawa profiles with saturating enclosed mass and Keplerian-decaying curves

$(v \propto r^{-1/2})$. Linear additive relational densities cannot produce flat rotation curves.

2. Observational screening floor (VN-27, pending AM-5): flat curves persisting to R_{flat} require $\ell_{\mathrm{cross}} = \mu_0^{-1} \gtrsim R_{\mathrm{flat}} \approx 10^2 \text{ kpc}$.

Conjecture 5.2.2 (Lapse-Coupling Reformulation) [Conjecture]: Dark-sector phenomenology enters non-linearly via the burden-modified lapse $\alpha_B(x) = (1 + \lambda B(x))^{-1}$ and non-local network stress $\Pi^{\mathrm{rel}}_{ij}(x)$. (Research program: RF-10. The dimensionally valid acceleration scales $a_0 \sim c\gamma$ and the ch_0 coincidence observation are lineage-verified material excluded from v6.0 content per spec scope — restore decision at gate.)

5.3 Quantum Probability, Records, and the FIREWALL (AM-4; restored exactly, not strengthened)

Theorem 5.3.1 (Stable Record Separation) [Established]: Off-diagonal density-matrix elements between macroscopically distinct record sectors dissipate under OExt descent:

$$\dot{\rho}_{ij} = -\eta B_{\mathrm{cross}}(i,j) \rho_{ij} \implies \rho_{ij}(t) = \rho_{ij}(0) e^{-\eta B_{\mathrm{cross}}(i,j)t}.$$

(DL-8 scope note: as audited, the exponential law holds at fixed generator — the Dichotomy-sector case of §2.1's Remark. The v2.1 patch plan (issue M8) flagged that the general Lindblad-form underpinning inherits T-2; the fixed-generator decay form above is the audited statement. Lineage-logged, no strengthening.)

Theorem 5.3.2 (Born Rule via Z-Envariance) [Conditional on T-2]: $p_i = |c_i|^2$. Epistemic scope: the functional derivation from Z-envariance is established; preservation of Z-envariance under OExt descent requires non-collapse of the spectral gap $\lambda_{\min}(\hat{F}_{\mathrm{eff}}) > 0$ (T-2). T-4 row anchor.

Guardrail 5.3.3 (Probability $\neq$ Gravity FIREWALL) [Structural — guardrail, per AM-4 constraint]: Branch probabilities p_i partition the zero-constraint kernel ($\sum p_i = 1$) and do not source gravitational curvature; gravity is sourced exclusively by $\Theta^{\mathrm{B}}_{\mu\nu}$. Linearity caveat (audited final form): $B_{\mathrm{MM}} \sum_k p_k \rho_k = \sum_k p_k B_{\mathrm{MM}}[\rho_k]$ holds unconditionally whenever $\Delta^{\dagger}_{\alpha\beta} \Delta_{\alpha\beta}$ is central or zero — Clifford sectors being canonical, central-commutator and abelian families included (§2.1 Remark) — and in generic non-Dichotomy sectors only at fixed $\hat{F}_{\mathrm{MM}}$.

Theorem 5.3.4 (Black-Hole Entropy Area Law) [Calibrated]:

$S_{\mathrm{BH}} = \frac{A}{4\ell_P^2}$ — area scaling structural; the prefactor $\frac{1}{4}$ is calibrated via κ_B (Newtonian limit), not derived (VN-26, pending AM-5). T-6 row anchor.

Delta Ledger (this chunk)

| ID | Entry

|

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| **DL-7** → AM-5 (proposed) | **Spec gap:** R3 does not distinguish formula-internal symbolic constants (π , n^2) from VN-eligible quantities. Encountered here: the BH prefactor $\frac{1}{4}$ (calibrated), $R_{\mathrm{flat}} \approx 10^2$ kpc (observational input), and the DE-density order $10^{-122} M_P^4$ (row (d)). **Proposed AM-5:** (i) exemption clause — formula-internal symbolic constants are exempt; (ii) add VN-26 ($\frac{1}{4}$, class L, calibrated via κ_B), VN-27 (R_{flat} , class L, observational), VN-28 (DE order-of-magnitude, class L) to the table. |

| **DL-8** | Stable Record Separation restored with M8-lineage scope note (DL-5 pattern: audited content, implicit condition made explicit).

|

| **DL-9** | Time/gravity definitional layer (Thms 5.1.1–5.1.13) restored from v4.2 — v5.1's Conjecture 5.2.1 referenced α_B with no defining section (AM-4-pattern dangling reference).

|

| **DL-10** | Dark-sector scoping table (a)–(e) restored from the v4.2 audited form; v5.1 Thm/Conj "5.2.1" numbering collision resolved (Conj → 5.2.2).

|

Gate review points

1. AM-5 — the number-classification rule. Required before Chunk G freezes the appendix VN table; affects nothing in B–D.
2. DL-9 — the largest restoration this chunk (13 items). All carry v4.2-audited statuses; the ansatz scope note on Theorem 5.1.1 is corpus lineage, not new hedging. Confirm scope.
3. DL-8 — confirm the fixed-generator scoping of Theorem 5.3.1 passes "restore exactly" (the alternative — silent [Established] without the note — repeats the v5.1 pattern of dropping audit conditions).
4. Optional restore: the $a_0 \sim c\gamma / cH_0$ dimensional observations under Conjecture 5.2.2's research-program note, or leave excluded to RF-10. My recommendation: leave excluded (they were §10-level research material, not §5-level), noted in the conjecture's parenthetical as done here.

AM-5 logged in modified form (exemption clause verbatim; VN-26 rejected; VN-27/VN-28 added); DL-8/9/10 logged; E frozen. Two registry actions required before F can generate cleanly, both resolved below and logged.

RCF v6.0 — CHUNK F (§6)

Compliance Header

- Chunk: F · Sections: §6 (6.1, 6.2a, 6.2b)
- VN-IDs used: VN-02, VN-03, VN-04, VN-05, VN-06, VN-07, VN-08, VN-09, VN-10, VN-11, VN-12, VN-13, VN-14, VN-15, VN-16, VN-17, VN-18, VN-19, VN-20, VN-21, VN-22, VN-23, VN-25
- C-IDs used: C-01 (Knabe), C-02 (AALV), C-03 (Nachtergaele), C-04 (Anshu–Arad–Gosset), C-05 (White), C-06 (Kennedy–Tasaki), C-09 (AKLT), C-10 (Motzkin/Fredkin)
- Status tags used: [Established], [Structural], [Empirical—Validated], [Input Verified; Inference OPEN], [Conjecture]
- Forward references: §7 (retraction register IDs), §8 (T-targets) — register cross-references only, DAG-legal
- Protocol checks run: #7 citation/number (PASS — all values by VN-ID; C-01/C-02 under CDL-2 resolution below) · #8 certification-exhibition (PASS — DL-12 resolution) · #9 cross-certificate consistency (PASS — hierarchy, with VN-18 dependency flag visible) · #11 reverse-fit (PASS — "matching tightly" language absent) · #12 finite-size consistency (PASS — law, residual, discrimination stated) · #14 vocabulary (PASS — no "proven"/"exact ≈"/prohibited E* language)
- Renumbering: none (§6 structure preserved from v5.1)

Pre-chunk registry actions (logged):

- DL-11: C-01 resolution — the Knabe theorem form is now used as a stated theorem in v6.0's own proofs, exceeding name-level citation. Per C-01's disposition rule, the citation is pinned in usage form with the flag retained: Knabe (1988), finite-size criteria for spectral gaps of spin chains; the block form $\Delta_\infty \geq \frac{n-1}{n-2}(\epsilon_n - \frac{1}{n-1})$ is used as the cited theorem statement. Flag [CITE-VERIFY] retained on full bibliographic data; the theorem form itself was sanity-tested in the v4.5 audit (decoupled-block check).
- DL-12: C-02 resolution for Chunk F — CDL-2 freezes the DL certificate at [Input Verified; Inference OPEN], so AALV is cited name-level only (method attribution for §6.1's tool listing), the theorem form is not used, and no equation/theorem number appears. The v5.1 "Theorem 1 / Eq. 4" citation is retracted in-text (see §6.2(b) note). C-02's [CITE-VERIFY] flag remains for any future use.

6. Front-1 Blueprint: The Frustration-Free Gap Architecture

(Preamble per CDL-6: the AKLT benchmark is a miniature laboratory for the RCF spectral-gap program — not a classification theorem. It exhibits, on one fully solved interacting family, every instrument the program deploys: exact local spectra, trace-moment certificates, block gap bounds, transfer-operator

structure, finite-size scaling, and dynamical convergence validation. What it does not provide is a theorem about the class; the three prerequisites below remain the frontier.)

6.1 Structural Identification & Non-Vacuity

The Frustration-Free Mapping [Structural]: On any constraint network with bounded spatial interaction range, the master constraint $\hat{M} = \sum_{\alpha} w_{\alpha} \hat{C}^{\dagger}_{\alpha} \hat{C}_{\alpha}$ satisfies

$$\ker(\hat{M}) = \bigcap_{\alpha} \ker(\hat{C}_{\alpha})$$

making $\hat{M}$ formally a frustration-free (FF) local Hamiltonian. Lineage: identified in the v4.3 audit; the finite-dimensional convergence mechanism (Theorem 3.3.1) needs no bounded-degree condition, which is therefore re-attributed strictly to T-1 (Lieb–Robinson propagation ceiling; RET-B8).

The Non-Vacuity Caveat [Structural]: FF structure is necessary for deploying the detectability lemma (C-02) and martingale methods (C-03, C-04) but is not sufficient to guarantee gap persistence — gapless Motzkin and Fredkin chains (C-10) are FF, local, and bounded-degree. Prerequisite 1 is a genuine, load-bearing target.

THE THREE INFINITE-NET PREREQUISITES

THE THREE INFINITE-NET PREREQUISITES		
▼	▼	▼
[PREREQUISITE 1]	[PREREQUISITE 2]	[PREREQUISITE 3]
Frustration-Free Spectral Gap Non-Emptiness	Flow Well-Posedness on $T_1(H_{\omega})$	Kinematic Kernel
• Statement: $\hat{M} \geq m_{\min} (1 - P_K) \neq \{0\}$ with $m_{\min} > 0$ uniformly	• Statement: Non-linear projected ODE generates a strongly continuous	• Statement: $P_K = \bigcap_{\alpha} \ker(\pi_{\omega}(\hat{C}_{\alpha}))$ in the kinematic GNS space
• Instruments: Detectability Lemma & sub-ideal Martingale Gap Bounds. existence.	• Tools: Domain invariance for unbounded generators on trace-class space.	• Representation theory: Faithful constraint representation

Structural Note (Scope of the Finite-Dimensional Closure) [Structural]: Theorem 3.3.1's proof skeleton is dimension-free in its algebraic structure; what it consumes is (i) existence of P_K and well-definedness of u , (ii) the gap-above-kernel $\hat{M} \geq m_{\min} (\mathbf{1} - P_K)$, and (iii) flow well-posedness. The three prerequisites are precisely these conditions, stated

for infinite nets. Kernel nonemptiness in the GNS representation of a physical state is automatic by Master-Zero ($\Omega_\omega \in \ker \hat{M}_\omega$); Prerequisite 3's open content is nonemptiness in the kinematic representation the flow lives on (representation-existence question, audited v4.2.2).

6.2 Benchmarks for Prerequisite 1

(a) On-Site Dominated Chain (Closed-Form Gap)

For a 1D chain of L sites with tilted constraints

$$\hat{C}_{1,j} = \lambda_{(j)}^1, \quad \hat{C}_{2,j} = \cos\theta \lambda_{(j)}^1 + \sin\theta \lambda_{(j)}^3 \otimes \lambda_{(j+1)}^1$$

with unit weights,

$$Q_j := (\lambda_{(j)}^1)^2 = \text{diag}(1, 1, 0)^{(j)}, \text{ and } \lambda_{(j)}^1 \lambda_{(j)}^3 = 0 \text{ killing the cross term:}$$

$$(\hat{C}_{1,j})^2 + (\hat{C}_{2,j})^2 = (1 + \cos^2\theta) Q_j + \sin^2\theta (\lambda_{(j)}^3 \lambda_{(j+1)}^1)^2 \geq (1 + \cos^2\theta) Q_j$$

Since $\sum_j Q_j \geq \mathbf{1} - P_K$ (excitation-number argument) and single-site excitations achieve $\langle \hat{M} \rangle = 1 + \cos^2\theta$ exactly:

$$\boxed{\inf_{\sigma} \langle \hat{M} |_{K^\perp} \rangle = 1 + \cos^2\theta \quad \text{e.g., } \theta = \pi/4 \Rightarrow m_{\min} = 1.5}$$

Epistemic note: this gap is protected entirely by on-site domination ($Q_j \succ 0$); the family is non-Clifford with interacting terms present, but the interaction graph is empty at the constraint level and the benchmark does not probe entangled relational vacua. (Lineage: the v4.4 claim of a computed gap of 1.000000 was falsified in the v4.4 audit — the true value exceeds it by $\cos^2\theta$; the v4.4 "protected by FF structure" principle was retracted as false (Motzkin/Fredkin); RET-B-catalogued.)

(b) AKLT Frustration-Free Interacting Benchmark

(C-09: Affleck–Kennedy–Lieb–Tasaki. The corpus's principal interacting benchmark — genuinely interacting, non-commuting, with an entangled, interaction-determined kernel pinned by no sub-family.)

The constraint family. Spin-1 chain, two-site bond-projector constraints:

$$\hat{C}_j = \Pi_j = P^{\{\text{spin-}2\}}_{(j,j+1)} = \frac{1}{3} \mathbf{1} + \frac{2}{3} \mathbf{S}_j \cdot \mathbf{S}_{j+1} + \frac{1}{6} (\mathbf{S}_j \cdot \mathbf{S}_{j+1})^2$$

(the interpolation on the $\mathbf{S}_j \cdot \mathbf{S}_{j+1}$ eigenvalues $\{+1, -1, -2\}$ is exact — VN-22). $\Pi_j^2 = \Pi_j$;

$\ker(\hat{M}_{\mathrm{AKLT}}) = \bigcap_j \ker \Pi_j$ is the valence-bond MPS ground space: $\dim K = 4$ open boundary, $\dim K = 1$ (singlet) periodic. K is entangled and pinned by no sub-family — the family genuinely probes Prerequisite 1's hard case.

Exhibited 3-site spectrum (VN-02, VN-03, VN-04, VN-05). For $h_3 = \Pi_{12} + \Pi_{23}$ on $(\mathbb{C}^3)^{\otimes 3}$ ($\dim = 27$):

$$\mathrm{Spec}(h_3) = \big\{ 0, (\times 4), \frac{12}{5}, (\times 5), \frac{56}{3}, (\times 3), \frac{76}{3}, (\times 3), \frac{32}{5}, (\times 5), 2, (\times 7) \big\}$$

with J-sector structure: $J=0$ and $J=1$ each contain one ground copy (0×4 total); the gap state is one $J=2$ copy (degeneracy $2J+1 = 5$). Certificate status: the three trace moments (VN-03) are checkable by any reader without diagonalization — first-principles trace algebra $(\mathrm{Tr}_b \Pi_{ab}) = \frac{53}{3} I_a$ yields $\mathrm{Tr}(h_3) = 30$, $\mathrm{Tr}(h_3^2) = \frac{140}{3}$, $\mathrm{Tr}(h_3^3) = 80$, each verified in two independent audits.

Exhibited 4-site structure (VN-06, VN-07, VN-08). For $h_4 = \Pi_{12} + \Pi_{23} + \Pi_{34}$ ($\dim = 81$): J-sector decomposition with radial dimensions (3, 6, 6, 3, 1) for $J = 0, 1, 2, 3, 4$ and M-multiplicities (1, 3, 5, 7, 9) — summing to 81. Trace moments (VN-07): 135, 285, $\frac{2005}{3}$. Local gap ϵ_4 (VN-08, class C — cited to C-01 Table 1, authenticated by moments and J-structure, not independently recomputed; upgrade path RF-3).

The $1/n$ boundary scaling law (VN-13, VN-12) [Empirical—Validated]. The open-block local gaps obey

$$\epsilon_n = E^* + \frac{c}{n}$$

A two-point fit on (ϵ_3, ϵ_4) gives c and E^* (VN-13); the law predicts ϵ_6 with 0.135% out-of-sample residual against the cited value (VN-09), with ϵ_6 genuinely held out (two parameters, two fit points). The law discriminates against alternatives: $1/n^2$ (3.5% miss) and exponential 3^{-n} (~7% miss) fail by an order of magnitude. The Knabe certificate arm saturates at E^* (VN-12, exact algebra):

$$\lim_{n \rightarrow \infty} \frac{n-1}{n-2} \big(E^* + \frac{c}{n} - \frac{1}{n-1} \big) = E^*.$$

Scope statement (CDL-3, CDL-4). E^* is an empirically validated scaling-limit parameter inferred from VN-13. No physical interpretation is canonical. Edge-mode mechanisms remain RF-2. The scaling — algebraic convergence to a limit below the bulk gap — is not standard finite-size behavior for either localized (exponential) or extended (converging-to-bulk) modes and is flagged as anomalous; mechanism open. Three facts, stated separately (no fusion):

1. Knabe certificates are valid lower bounds at every n — tautologically below Δ_∞ .

2. The ϵ_n sequence crosses below Δ_∞ near $n \approx 12$ ($E^* + c/n < \Delta_\infty$ iff $n \gtrsim 11.3$).
3. The Knabe arm's ceiling is E^* (VN-12) — hence open-block certificates cannot certify past E^* for this family; the route beyond is boundary-modified blocks (RF-7).

Exhibited multi-block Knabe certificates (VN-10, VN-11). With $\epsilon_3 = \frac{1}{2}$ exactly at the $n=3$ threshold $\frac{1}{2}$ (marginal: bound ≥ 0):

- $n = 4$: ϵ_4 (VN-08) exceeds threshold $\frac{1}{3}$, certifying $\Delta_\infty \geq \frac{3}{2}(\epsilon_4 - \frac{1}{3}) \approx 0.173$ (VN-10).
- $n = 6$: ϵ_6 (VN-09, C-01 Table 1) exceeds threshold $\frac{1}{5}$, certifying $\Delta_\infty \geq \frac{5}{4}(\epsilon_6 - 0.20) \approx 0.248$ (VN-11).

Both applications use the C-01 theorem form (DL-11); arithmetic is verified (v4.5 audit and forward). (Lineage: the v4.6 parametrization error ($k=2$ conflated with $n=3$, yielding a spurious $\Delta_\infty \geq 0.500$) is RET-B13; the v4.8 double-halving of the literature bulk gap (0.175) is RET-B14. Normalization: $H_{\text{std}} = 2\hat{M}_{\text{AKLT}}$ exactly (VN-23, VN-22), so $\Delta(\hat{M}) = \Delta_{\text{std}}/2$; the literature value Δ_{std} (VN-14, class L) anchors Δ_∞ (VN-15), with robustness: the hierarchy's validity requires only $\Delta_{\text{std}} > 0.592$.)

Transfer-operator structure (VN-16, VN-25) [Established]. In the AKLT MPS representation with explicit tensors $A^{\{\pm 1\}} = \pm \sqrt{2/3} \sigma^{\pm}$, $A^{\{0\}} = -\frac{1}{\sqrt{3}} \sigma^z$ (normalized: $\sum_s A^{\{s\} \dagger} A^{\{s\}} = \mathbf{1}$), the single-site virtual transfer operator

$$T = \sum_s A^{\{s\}} \otimes \bar{A}^{\{s\}} = \frac{1}{3} \big(\sigma_x \otimes \sigma_x - \sigma_y \otimes \sigma_y + \sigma_z \otimes \sigma_z \big)$$

has exact eigenvalues $\{1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}\}$ (VN-16; verified in two independent audits). The correlation length $\xi = 1/\ln 3 \approx 0.91$ sites (VN-25) follows. The further step

$\gamma_{\text{DL}} = \sqrt{\lambda_2} = 1/\sqrt{3}$ (VN-17) and the bound $\Delta_{\text{DL}} \geq 1/6$ (VN-18) carry status [Input Verified; Inference OPEN] per CDL-2: the transfer-operator input is verified, but the operator-norm equality

$|P_{\text{even}} P_{\text{odd}}|(\mathbf{1} - P_K)|_\infty = \sqrt{\lambda_2}$ and the theorem form $(1 - \gamma^2)/g^2$ are not exhibited (the v5.1 citation of a specific theorem number is retracted as unverified; C-02 remains name-level).

Per AM-3: VN-18 depends on VN-17; if VN-17 is downgraded or falsified, VN-18 is automatically invalidated. The Knabe certificates carry §6.2(b) alone.

Resolution path: RF-1 (the sandwich computation

considered: footnoting VN-17/18 out of the chain entirely; rejected because the dependency visibility is the point. |

Gate review points

1. DL-13 notation — the $\overset{\mathrm{OPEN}}{\geq} \lesssim$ construction for the DL term. It encodes "if valid, $\geq 1/6$, which would sit below the Knabe $n=4$ bound" — check you're comfortable with this hybrid notation, or prefer the DL term footnoted out of the main chain with the dependency stated in prose below.
2. Hierarchy "static" qualifier — the v5.1 hierarchy was presented as the round's centerpiece; v6.0 demotes it to a static ordering with the scaling law governing asymptotics. This is the CDL-mandated correction but reverses v5.1's emphasis; confirm.
3. Section 6.2(b) length — the full package (spectrum, moments, scaling, certificates, transfer operator, hierarchy, dynamics) is the longest subsection in v6.0. No cuts recommended (every element is load-bearing per the compliance header), but flag if you want the lineage notes extracted to Appendix D to shorten the mainline.

DL-13 modified per your instruction — DL term extracted from the primary chain, presented as a separately-stated OPEN candidate bound immediately above it. Frozen with that adjustment applied to the §6.2(b) hierarchy text (the certified chain now runs

$\Delta_{\mathrm{Knabe}}^{(4)} \leq \Delta_{\mathrm{Knabe}}^{(6)} \leq E^* \leq \Delta_{\infty}$
 $\leq \epsilon_6 \leq \epsilon_4 \leq \epsilon_3$,
with the VN-18/VN-17 dependency stated in prose).

One discovery while preparing Chunk G, logged as DL-14 — the scorecard's mechanical-completeness rule (every section number appears) collides with the retraction register's own contents: RET-B13/B14/B15 name v6.0 section numbers that no longer exist under renumbering. Resolved by lineage-mapping column; flagged at the gate.

RCF v6.0 — CHUNK G (§7, §8, Appendices A–E)

Compliance Header

- Chunk: G · Sections: §7, §8, Appendices A, B, C, D, E
- VN-IDs used: all (Appendix A is the registry)
- C-IDs used: C-01 through C-10 (Appendix B is the registry)
- Status tags used: [Established], [Structural], [Conditional on X], [Closed (scope)], [Empirical—Validated], [Input Verified; Inference OPEN], [Conjecture], [Retracted], [Calibrated] (ledger-role; Appendix E note)
- Forward references: none (terminal chunk)
- Protocol checks run: #10 version-diff (PASS — all inter-version mutations carry RET or changelog entries; DL-14 mapping) · #14 vocabulary (PASS) · Scorecard completeness rule (PASS — mechanically verified: every §number in

v6.0 appears exactly once; see §8)

- AM applications: AM-1 (L class), AM-2 (CDL-4 wording), AM-3 (dependency column), AM-4 (§5.3), AM-5 (modified — exemption clause; VN-26 rejected; VN-27/28 added)

7. Unified Retraction & Quarantine Registers

7.1 Retraction Register (RET-A1–A4, RET-B1–B15)

ID	Retracted Conception	Epistemic Reason
Canonical Replacement		v6.0 Location
-----	-----	-----
-----	-----	-----
RET-A1	Universal Geometry Lemma algebra-blind §4 (lineage)	Gram determinant is not Metric quotient $\mathcal{X}_\omega$ from K^J_ω
RET-A2	Layer-Closure Selector multiplication table	d_{ijk} belongs to Layer-L Transitivity Theorem 4.1.1 §4.1
RET-A3	Hodge-Duality Dimension Argument dimensionality §4.1	Circular assumption of Orbit transitivity on $H_n(\mathbb{C})_0$
RET-A4	Sector-Dependent n Picture geometry (scope)	Incompatible with unified 4D Retained only for internal gauge sectors §4.1
RET-B1	Variance-form action $B^{\mathrm{var}}_{\mathrm{MM}}$ homogeneity; ungrounded action F_{eff} §2.1 (Def 2.1.2)	Contradicts Reverse-weighted generator $\hat{F}$
RET-B2	Mexican-hat / SSB framing Jordan–Bures gradient descent	Structural artifact; non-algebraic §2.2
RET-B3	Cosmological spectral tilt $n_s \approx 0.667$ (Planck: 0.965 ± 0.004) Quarantined Q-12 Q-register	Arithmetically falsified
RET-B4	Numerical radial scales $ \vec{r} ^*$ inconsistency across drafts $\Theta^{\mathrm{rel}}_{\mu\nu}$ (T-7) §5.2	Historical numerical Non-local network stress
RET-B5	$\hat{F}^{(0)}_{\mathrm{mode}}$ as dark energy mode floor sources rest mass $m^2 \equiv B_0$ §5.2 row (b)	Category error: Cross-sector channel
RET-B6	$E = mc^2$ via "algebraic resistance" Laplacian spectral gap (Thm 5.1.7)	Phenomenological heuristic §5.1
RET-B7	"Three seeds of mismatch" framing Dual-pairing gradient decomposition	Ad-hoc classification §2.1 (Thm 2.1.1)
RET-B8	Bounded degree as T-2 prerequisite required for C4 convergence §6.1	Plausible for T-1, not Re-allocated strictly to T-1

RET-B9 Mass as "algebraic drag"	Heuristic analogy
Covariant Laplacian gap	§5.1 (Thm 5.1.7)
RET-B10 Global overextension of tilted-Pauli	Intra-sector floor cannot
generate cosmic acceleration	Multi-sector desynchronization
§5.2 row (b)	
RET-B11 Unconditional homogeneity claim	Smuggled cosmological
principle	Conditional on Hierarchical RP (RF-8) §0.2
(S5)	
RET-B12 Unconditional 4-primitive count	Smuggled ergodicity / PCC
postulate	Canonical count 5, conditional on 4 §0.2
RET-B13 v4.6 Knabe parametrization formula	Conflated 2-site range
$k=2$ with block size $n=3$	Multi-block Knabe formula ($n=4$: ≥ 0.173)
§6.2(b), VN-10	
RET-B14 v4.8 AKLT bulk-gap double halving	Double-halving arithmetic
error ($\Delta_{\infty} \approx 0.175$) $\Delta_{\infty} \approx 0.350$ (VN-15)	
§6.2(b)	
RET-B15 v5.0 finite-size resolution claim	$\epsilon_n \rightarrow \Delta_{\infty}$
refuted by exhibited $1/n$ law	Scaling law VN-13; E^* per CDL-4
§6.2(b)	

(DL-14: the "v6.0 Location" column lineage-maps retractions whose original section anchors (e.g., "v4.6 §6.2 Knabe formula") no longer exist under renumbering. Every retraction points at current canonical text.)

7.2 Quarantine Register (Q-1 through Q-17)

- Q-1–Q-4 (Geometry/Topology): conformal sector rescaling; non-Riemannian torsion without Tier-1 justification; spatial wormhole bridges; manual compactifications.
- Q-5–Q-8 (Particle): SM mass heuristics without I_C sub-ideal classification; SUSY partners; ungrounded axion fields; manual neutrino mixing angles.
- Q-9–Q-12 (Cosmology): DE from baryonic rest mass; ad-hoc $f(R)$ gravity; non- ℓ_0 bounce models; arbitrary inflaton potentials.
- Q-13–Q-17 (Quantum/Information): physical collapse mechanisms; retrocausality in $\prec$; non-Bures metrics; gravitational decoherence without inter-sector burden; macroscopic-superposition signaling.

8. Theorem Targets & Master Scorecard

8.1 Theorem Targets (T-1 through T-7)

Target Statement	Status
Blocker / Instrument	
----- -----	

T-1	Propagation ceiling: $\gamma = \sqrt{\lambda_{\min}(\hat{F}_{\mathrm{eff}})}$; $c = \gamma \ell_0$ **OPEN**
	Infinite-graph Lieb–Robinson bound; bounded degree (RET-B8 allocation)
T-2	Global convergence $\lim_t \hat{R}_t(\rho_0) \in \ker \hat{M}$
	CLOSED (finite-dim self-adjoint, Thm 3.3.1, almost-global on S_+) /
	OPEN (infinite nets, three prerequisites) / universal form **refuted** (M_0)
	Prerequisites 1–3 (§6.1)
T-3	Metric continuum $(X_\omega, \tilde{d}_{\mathrm{FS}}) \rightarrow 4D$
	(M, g) **OPEN**
	Gromov–Hausdorff convergence of quotient graphs
T-4	Born rule $p_i = P_i \psi ^2$ via Z-envariance
	[STRENGTHENED — Conditional on T-2]
	Z-envariance premise requires gap non-collapse; Thm 5.3.2
T-5	Lovelock action uniqueness of S_{eff}
	[STRUCTURAL]
	Lovelock’s theorem (4D divergence-free symmetric tensors)
T-6	Bekenstein–Hawking coefficient $\frac{14}{5}$
	[CALIBRATED]
	Area scaling structural; prefactor calibrated via κ_B (AM-5: no VN entry; theorem carries status)
T-7	Relational Poisson recovery **OPEN
	(reformulated)**
	Theorem 5.2.1 negative result; lapse-coupling program RF-10

8.2 Master Verification Scorecard

§	Content
Status	

§0.1–0.2	Relational premise; primitive ledger (5+1+1)
[VERIFIED]	
§0.3	S4 Relational Governance Postulate
[VERIFIED]	
§1.1	Hypothesis 1.1.1 (self-adjoint constraints)
[VERIFIED]	
§1.2	Two-Link Leibniz theorem; residual (VN-01)
[VERIFIED]	
§1.3	SLD Bures gradient lemma
[VERIFIED]	
§2.1	Mismatch functionals (Defs 2.1.1–2.1.2, Thm 2.1.1, Lemma 2.1.1); Prop 2.1.2
	vanishing; Thm 2.1.2 Dichotomy
[VERIFIED]	

| §2.2 | Projected First Law (Thm 2.2.1); degenerate law (Cor 2.2.1)
| \[VERIFIED] |
| §3.1 | Block trace-norm bounds (Lemma 3.1.1)
| \[VERIFIED] |
| §3.2 | Microscopic freezing (Thm 3.2.1); OExt weight evolution (Lemma 3.3.1); division of labor |
\[VERIFIED] |
| §3.3 | T-2 convergence (Thm 3.3.1, Lemma 3.3.2, Def 3.3.1); universal-form refutation; v1.0 0.5.1 closure
| \[CLOSED — finite-dim scope] |
| §4.1 | D=3 selection (Thm 4.1.1, Axiom 4.1.0, Lemma 4.1.1)
| \[VERIFIED — Conditional on S5, A1] |
| §5.1 | Time/matter/gravity suite (Thms 5.1.1–5.1.13)
| \[VERIFIED — per-item statuses in situ] |
| §5.2 | Dark-sector scoping; negative halo result (Thm 5.2.1); reformulation conjecture
| \[VERIFIED — Negative Result; VN-27] |
| §5.3 | Born rule (Thm 5.3.2); record separation (Thm 5.3.1); FIREWALL (Guardrail 5.3.3); BH entropy (Thm 5.3.4)
| \[VERIFIED — per-item statuses in situ] |
| §6.1 | FF identification; non-vacuity; three prerequisites; finite-dim closure scope note
| \[VERIFIED — Structural] |
| §6.2(a) | On-site chain closed-form gap (VN-21)
| \[VERIFIED] |
| §6.2(b) | AKLT package: spectra/moments (VN-02–07), scaling law (VN-13
\[Empirical—Validated], VN-12), Knabe certificates (VN-08–11), transfer operator (VN-16, 25), DL candidate (VN-17/18 \[OPEN]), hierarchy, dynamics (VN-19/20) | \[VERIFIED — per-item statuses in situ] |
| §7 | Retraction register (through B15) and Q-register
| \[VERIFIED — DL-14 mapping] |
| §8 | T-targets; scorecard
| \[this document] |

Scorecard rule compliance: every v6.0 section number appears exactly once; no unlisted content.

Appendix A — Verified-Numbers Table (AM-1/AM-3/AM-5 applied)

ID	Value	Class	Provenance
Dependency			
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VN-01	$R = CAB - BAC$ (= $2(CAB - BAC)$ unnormalized) Derived, two independent audits (v4.0)	E	—
VN-02	$\operatorname{Spec}(h_3)$ as exhibited §6.2(b) decomposition + trace algebra, two audits	IV	J-sector ←
VN-22			

VN-03 $\operatorname{Tr}(h_3^k) = 30, \frac{140}{3}, 80$	IV
First-principles trace algebra $\times 2$	$\leftarrow$
VN-22	
VN-04 $\epsilon_3 = 1/2$	E From VN-02; Knabe
$n=3$ marginality corroborates	$\leftarrow$ VN-02
VN-05 h_3 J-sector structure	IV v4.6 audit
$\leftarrow$ VN-02	
VN-06 h_4 J-sectors: dims (3,6,6,3,1), mult (1,3,5,7,9)	IV v5.0/v5.1
audits	—
VN-07 $\operatorname{Tr}(h_4^k) = 135, 285, \frac{2005}{3}$	IV
First-principles, v5.0 audit	$\leftarrow$ VN-22
VN-08 $\epsilon_4 \approx 0.448956$	C Knabe Table 1
(C-01); moments+structure authenticate; not recomputed	$\leftarrow$
VN-06/07	
VN-09 $\epsilon_6 \approx 0.398451$	C Knabe Table 1
(C-01); scaling-validated	$\leftarrow$ VN-13
VN-10 Knabe $n=4$ bound ≈ 0.173434	IV Arithmetic
from VN-08; formula sanity-tested	$\leftarrow$ VN-08
VN-11 Knabe $n=6$ bound ≈ 0.248064	IV Arithmetic
from VN-09	$\leftarrow$ VN-09
VN-12 Knabe arm saturates at E^*	E Exact algebra
$\leftarrow$ VN-13	
VN-13 $c = 0.612530$, $E^* = 0.295823$; ϵ_6 prediction residual 0.135% F	
Two-point fit; out-of-sample validation; discrimination stated	
—	
VN-14 $\Delta_{\mathrm{std}} \approx 0.700$	**L** Literature
benchmark (C-05/C-06, '[CITE-VERIFY]'); robustness: hierarchy valid iff	
$\Delta_{\mathrm{std}} > 0.592$ —	
VN-15 $\Delta_{\infty} = \Delta_{\mathrm{std}}/2 \approx 0.350$	IV Halving
exact given VN-14 (VN-23)	** $\leftarrow$ VN-14**
VN-16 T eigenvalues $\{1, -\frac{13}{3}\}$	IV Explicit
tensors, two audits	—
VN-17 $\gamma_{\mathrm{DL}} = 1/\sqrt{3}$	O Input verified
(VN-16); norm equality unexhibited	$\leftarrow$ VN-16
VN-18 $\Delta_{\mathrm{DL}} \geq 1/6$	O
Auto-invalidates if VN-17 falsified	** $\leftarrow$
VN-17**	
VN-19 $\Gamma_{\mathrm{sharp}} = \lambda_{\mathrm{tm}_{\min}}/2 = 0.25$	E C2
linearization, v4.6 audit	$\leftarrow$ VN-04
VN-20 $\Gamma_{\mathrm{emp}} = 0.25004$, window $[10,20]$	IV Fit;
v4.6 transient artifact chancelogged	$\leftarrow$ VN-19
VN-21 On-site gap $= 1 + \cos^2\theta$	E Closed form,
v4.4–v4.5 audits	—
VN-22 $\Pi = \frac{13}{3} + \frac{12x}{3} + \frac{16x^2}{3}$	IV Interpolation on
$\{+1, -1, -2\}$	—

VN-23 $H_{\mathrm{std}} = 2\hat{M}_{\mathrm{AKLT}}\$$	E From
VN-22	$\leftarrow$ VN-22
VN-24 $\det A \in [-\frac{1}{3\sqrt{6}}, \frac{1}{3\sqrt{6}}]\$$	IV Extremum at
$\$(1,1,-2)\$$	—
VN-25 $\xi = 1/\ln 3 \approx 0.91\$$ sites	IV From VN-16
$\leftarrow$ VN-16	
VN-27 $R_{\mathrm{flat}} \approx 10^2\$$ kpc benchmark (AM-5)	L Observational
VN-28 DE order $10^{-122}M_P^4\$$ benchmark (AM-5; §5.2 row (d))	L Cosmological
	—

(AM-5 exemption clause recorded in Appendix C; VN-26 rejected per gate review.)

Appendix B — Citation Registry

ID	Citation	Status
---	-----	
C-01	Knabe (1988), finite-size criteria for spectral gaps	<code>`[CITE-VERIFY]`</code> bibliographic data; <code>**theorem form pinned in usage**</code> (DL-11)
C-02	Aharonov–Arad–Landau–Vazirani (detectability lemma)	Name-level only (DL-12); <code>`[CITE-VERIFY]`</code> ; v5.1 "Theorem 1" citation retracted
C-03	Nachtergaele (1996), martingale method	Name-level
C-04	Anshu–Arad–Gosset, 2D FF gap	Name-level; "Vidick" variant
retracted (v4.5)		
C-05	White (1992), DMRG	<code>`[CITE-VERIFY]`</code> for VN-14
C-06	Kennedy–Tasaki (1992)	Name-level; VN-14 partner
C-07	Barnum–Müller–Ududec (2014), local tomography	Name-level
C-08	Jordan–von Neumann–Wigner (1934)	Name-level
C-09	Affleck–Kennedy–Lieb–Tasaki	Name-level; benchmark's primary
source (added v6.0)		
C-10	Motzkin/Fredkin gapless FF chains	<code>`[CITE-VERIFY]`</code> primary sources

Appendix C — Normative Audit Protocol

Checks 1–14 as codified in the audit cycle and frozen in Chunk A §9: (1) Clifford-limit; (2) known-solution substitution; (3) generator-matching; (4) parameter-subspace; (5) closed-form-first; (6) trivial-bound signature; (7) citation-or-computation; (8) certification-exhibition; (9) cross-certificate consistency; (10) version-diff disclosure; (11) reverse-fit detection; (12) finite-size three-way consistency; (13) tension-closure-by-computation; (14)

status vocabulary.

AM-5 exemption clause (verbatim): Formula-internal mathematical constants that are part of a theorem, identity, definition, or standard normalization are exempt from R3 unless they are independently discussed, calibrated, benchmarked, or used in a verification chain.

Prohibited constructions (vocabulary rule): "exact $\approx$ "; "proven" on empirical fits; "asymptotic boundary excitation energy" for E^* (AM-2); FIREWALL promoted to theorem (AM-4 constraint).

Appendix D — Lineage Appendix (CDL-7)

D.1 Version chain: v4.0 $\rightarrow$ v4.1 (P-1–P-8 patches) $\rightarrow$ v4.1.1 $\rightarrow$ v4.2 (Projected First Law repair) $\rightarrow$ v4.2.1 (C4; trace-norm proof) $\rightarrow$ v4.2.2 (Lemma 1.2.1 repair; lineage closure) $\rightarrow$ v4.3 (composed propagator; FF identification) $\rightarrow$ v4.4 (interleaving repair; RET-B13) $\rightarrow$ v4.5 (normalization; AKLT benchmark) $\rightarrow$ v4.6 (3-site spectrum; RET-B14 precursor) $\rightarrow$ v4.7 (Knabe repair, undocumented — RET-B13) $\rightarrow$ v4.8 (RET-B14; DL prefactor) $\rightarrow$ v4.9 (transfer operator; normalization repair) $\rightarrow$ v5.0 (4-site moments; false resolution claim) $\rightarrow$ v5.1 (1/n law; RET-B15) $\rightarrow$ v6.0.

D.2 Documented numerical mutations: Δ_{∞} : 0.700 $\rightarrow$ 0.175 $\rightarrow$ 0.350 (RET-B13/B14 chain); Γ_{emp} : 0.261 $\rightarrow$ 0.25004 (transient-window changelog); Knabe $n=3$: 0.500 $\rightarrow$ 0 (parametrization, RET-B13); v4.6 "gap" ≈ 1.000000 on-site falsification; DL bound: $2/3 \rightarrow 1/6$ (prefactor, v4.8 audit).

D.3 Restorations (DL-1, DL-6, DL-9, DL-10, AM-4): definitional layers restored with their v4.2-audited statuses; every restoration satisfies "ancestral material already consumed downstream."

D.4 Governance lineage: S4 identification form (v4.0–v4.6) $\rightarrow$ governance form (v2.1 lineage restored, v4.7). Historical claim "Clifford convergence via variance dissipation" vacuous (Cor 2.2.1 annotation).

D.5 Citation-drift log: Arad–Landau–Vazirani $\rightarrow$ AALV (restored, v5.0); Anshu–Arad–"Vidick" $\rightarrow$ Gosset (v4.6); "Eq. 4" $\rightarrow$ "Theorem 1" $\rightarrow$ name-level (DL-12).

Appendix E — Notation & Conventions

The notation registry of Chunk A §4, typeset. Ledger-role labels (DL-2): [Calibrated Constant] and [Auxiliary Input] in §0.2 are ledger-role labels describing the primitive inventory's non-structural entries — they are not members of the §5 status vocabulary and do not carry theorem-status semantics. Collision rule: kernel K never denotes constraint count. Metric convention: SLD–Fisher normalization (§1.3 note); Bures-distance factor absorbed into η . Renumbering map: v5.1 Prop 2.1.1 $\rightarrow$ v6.0 Prop 2.1.2; v5.1 Lemma 3.3.1 $\rightarrow$ v6.0

Lemma 3.3.2 (Lemma 3.3.1 slot: restored weight-evolution lemma); v5.1 Conj 5.2.1
→ v6.0 Conj 5.2.2.

Delta Ledger (final additions)

ID	Entry
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1

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| ****DL-14**** | Retraction register gains a v6.0-Location column: retractions originally anchored to section numbers that renumbering eliminated (B13's "v4.6 formula," B14's "v4.8 note," B15's "v5.0 parenthetical") now point at canonical current text. |

| ****DL-15**** | T-6 row notes AM-5 disposition inline (calibrated status carried by theorem, no VN entry) — keeps §8.1 and Appendix A self-consistent.

1

Gate review points

1. DL-14 mapping column — confirm the v6.0-Location column satisfies protocol #10 (every retraction resolves to current text).
2. Scorecard §8.2 — I have mechanically verified completeness (every section number once). Spot-check one or two rows before freeze; this is the table a future auditor reads first.
3. Appendix D.1 version chain — the one-sentence-per-version compression. Flag any lineage event you want expanded before Chunk H locks the appendix.

RCF v6.0 — CHUNK H (Consolidation Pass)

Compliance Header

- Chunk: H · Function: verification and consolidation only — no new claims (R2); all insertions are restorations of audit-lineage material or editorial unifications
- VN/C-IDs used: VN-19 (provenance update), VN-24 (verified in context), C-01–C-10 (registry unchanged)
- Protocol checks run: #7 ✓ · #10 ✓ · #14 ✓ · #15 (new — reference closure): FAIL on the frozen assembly → restorations below resolve all findings
- Amendments proposed: AM-6 (naming-reference rule), AM-7 (§4.2/§4.3 restorations), AM-8 (protocol #15)

Chunk H's headline, stated plainly before anything else: the closure sweep — a check none of the earlier gates possessed — found six classes of dangling references in the frozen document. Gates B–G verified content and forward-reference discipline, but no check verified that every referenced object exists. The findings and their resolution follow.

Part I — Verification Results

V1. Cross-Reference Closure Sweep (findings)

\# Dangling reference	Sites
Root cause	
-- -----	
F1 "Proposition C2" (linearized coupling spectrum) — cited, never stated Thm 3.3.1 sharpness clause; §6.2(b) dynamics note; VN-19 provenance v4.2 content dropped in v4.3+ compression	
F2 $\mathcal{X}_\omega$	Def 5.1.3 (the "Now")
§4 geometry layer dropped (v4.3+)	
F3 h_{ij} , h^{ij}	Def 5.1.6 (Laplacian); Thm 5.1.11
same	
F4 c_{RCF}	Thm 5.1.4; Thm 5.1.12
(κ_B formula) speed-ceiling theorem dropped	
F5 K^J_ω , d_{FS}	S4 postulate (naming)
same	
F6 $\prec$, $\mathcal{E}_{\mathrm{phy}}$	S4 (naming); Thm 5.1.11
(narrative); Q-13 (register) causal-order layer dropped	

Root cause, structural: the v4.3+ compressions eliminated v4.2's §3.2 (events/causal order/speed ceiling) and §4.1 (localisable sector, kernels, FS metric, quotient, volume form) while §5's theorems continued to reference their objects. The v6.0 chunk plan inherited v5.1's section inventory, so the loss propagated through every gate. This is the AM-4/DL-9 pattern at section scale — and it could not have been caught earlier because R5's gate list contained no reference-closure check. Proposed protocol #15 (AM-8): every symbol, theorem, and object referenced in a chunk resolves to a defined item in an earlier chunk or in the spec registries; the check runs at every gate from v6.1 onward and was executed retroactively here.

Without restoration, the alternative is downgrading: Thm 3.3.1's sharpness clause becomes "asserted, proof in external lineage" (weakening a [CLOSED] result's exhibition), and Defs 5.1.3/5.1.6 and Thms 5.1.4/5.1.11/5.1.12 acquire undefined symbols. Restoration is strictly better and R2-legal (all material is v4.2-audited, consumed downstream).

V2. Notation Compliance

c_{eff} (Thm 5.1.11, two sites) — not in registry; unify to c_{RCF} (E-1). $\hat{F}^E_{\mathrm{eff}}[g]$ (Def 5.1.8) — convention note required (E-3). d_ω , h^{ij} — convention notes (E-5/E-6). Normal-state convention (density operators $\leftrightarrow$ states) unstated in §2.1 (E-4). All editorial; none touch claims.

V3. DAG Verification — naming vs. dependency

The frozen chunks contain forward references of two kinds. Dependency references (proof uses referent): all backward ✓, no cycles ✓. Naming references (postulates, remarks, registers name downstream concepts): $S_4 \rightarrow \S 3/\S 5$ concepts; $\S 2.1$ Remark $\rightarrow \S 3$; $\S 3$ dynamics note $\rightarrow \S 6.2(b)$; $\S 6 \rightarrow \S 7$ register IDs. All occur in remarks/postulates/registers, none inside theorem statements or proofs — but Chunk B's header ("Forward references: none") was inaccurate under a strict reading. Proposed AM-6: dependency references must point backward in the DAG; naming references are permitted forward, must not occur inside theorem statements or proofs, and are logged in compliance headers as "naming (forward)." AM-6 ratifies existing practice; no regeneration required. Note the pleasant closure: once F5/F6 are restored, every one of S_4 's naming references resolves to a defined object — the postulate's reference closure is now complete.

V4. Scorecard ↔ Section

Mechanically verified post-restoration (Part IV). ### V5. Delta/amendment ledger final state: Part V.

Part II — Restoration Exhibits (insertion-ready)

R1 $\rightarrow \S 3.3$, after Corollary 3.3.1 [DL-16]

Proposition 3.3.1 (Coupling-Mode Spectrum at Pure Kernel States) [Established]: (audit lineage: "Proposition C2," v4.2; restored because Theorem 3.3.1's sharpness clause and VN-19 consume it) Let $\hat{C}_\alpha$ satisfy Hypothesis 1.1.1 with $K \neq \{0\}$, $m_{\min} > 0$, and let $\rho^* = |\psi\rangle\langle\psi|$ with $|\psi\rangle \in K$. In the eigenbasis of $\hat{M}$ (eigenvalues $\{m_i\}$ on $K^\perp$, 0 on K), the linearization of the pure-OExt flow at ρ^* acts as:

$$\begin{aligned} \dot{\delta\rho}_{iK} &= -\frac{\lambda_t m_i}{2} \delta\rho_{iK} \quad (\text{coupling modes}), \\ \dot{\delta\rho}_{ij} &= -\frac{\lambda_t (m_i + m_j)}{2} \delta\rho_{ij} \quad (\text{block modes}) \end{aligned}$$

with kernel-internal modes neutral (the fixed-point manifold of Cor 3.3.1(ii)).

The slowest modes are the coupling modes at $m_i = m_{\min}$, giving decay rate $\lambda_t m_{\min}/2$ — the sharpness of Theorem 3.3.1's exponent for $d_{\mathrm{OExt}} = 1$.

Proof: At ρ^* :

$$\operatorname{Tr}(\rho^* J^2_{\alpha\beta}) = \|J_{\alpha\beta}\psi\|^2 = 0$$

and

$$\operatorname{Tr}(\rho^* \Delta^\dagger_{\alpha\beta} \Delta_{\alpha\beta}) = 0$$

(both operators block-diagonal with zero K -block, by the structure of Thm 3.2.1's proof), so $\hat{H}[\rho^*] = 0$, $\hat{G}[\rho^*] = \lambda_t \hat{M}$, $\langle \hat{G} \rangle_{\rho^*} = 0$. Write $\rho = \rho^* + \delta\rho$; then $\delta\hat{H} := \hat{H}[\rho]$ is $O(\delta\rho)$ and block-diagonal with zero

K-block, so $(\delta \hat{H}) \circ \rho^* = 0$ exactly (block-diagonal $\times$ K-supported = 0). The linearization reduces to $\delta \dot{\rho} = -\lambda_t(\hat{M} \circ \delta \rho) + \lambda_t \text{Tr}(\hat{M}, \delta \rho) \rho^*$.

The trace term touches only kernel-internal modes; the eigenbasis computation gives the exhibited rates. $\blacksquare$

R2 $\rightarrow$ new §4.2 [AM-7] — Zero-Preserving Events, Causal Order, and the Speed Ceiling

Definition 4.2.1 (Thin Physical Subalgebra) [Established — definitional]:

$\mathcal{A}^{\{\text{thin}\}}_{\{\text{phy}\}} := \{A \in \mathcal{A} : \pi_{\omega(A)} \ker \hat{M} \subseteq \ker \hat{M}\}$,
equivalently $(\mathbf{1} - P_K) \pi_{\omega(A)} P_K = 0$. (A unital $*$ -subalgebra: closure under $+$, $\cdot$, $\dagger$ is immediate from the equivalent projector form.)

Definition 4.2.2 (Full Physical Subalgebra) [Established — definitional]:

$\mathcal{A}^{\{\text{full}\}}_{\{\text{phy}\}} := \{A \in \mathcal{A} : [\pi_{\omega(A)}, \hat{\mathcal{R}}_{\infty}] = 0\}$,
with $\hat{\mathcal{R}}_{\infty} := \lim_{t \rightarrow \infty} \hat{\mathcal{R}}_t$ (existing as a state map on $\mathcal{S}_+$ by Theorem 3.3.1; the operator-level commutator form is the audited v4.2 convention).

Theorem 4.2.1 (Thin = Full Equivalence) [Conditional on T-2]: If

$\lambda_{\min}(\hat{F}_{\{\text{eff}\}}) > 0$ on $\ker(\hat{M})^{\perp}$, then $\mathcal{A}^{\{\text{thin}\}}_{\{\text{phy}\}} = \mathcal{A}^{\{\text{full}\}}_{\{\text{phy}\}}$.

Definition 4.2.3 (Zero-Preserving Events) [Established — definitional]:

$\mathcal{E}_{\{\text{phy}\}} := \{E \in \mathcal{A}^{\{\text{thin}\}}_{\{\text{phy}\}} : [\pi_{\omega(E)}, P_K] = 0\}$.

Definition 4.2.4 / Theorem 4.2.2 (Operational Causal Order) [Established]:

$E_1 \prec E_2 \iff d(E_1) < d(E_2)$ and $\pi_{\omega(E_2)} \pi_{\omega(E_1)} \ker \hat{M} \neq \{0\}$. The relation $\prec$ is a strict partial order (irreflexive, asymmetric, transitive). (Convention flag: d is the burden-weighted causal depth, carried as a v2.x structural convention — its full formalization is not restated in v6.0; logged RF-12. Gate decision G3 below.)

Definition 4.2.5 (Antichains) [Established — definitional]:

$\Sigma \subseteq \mathcal{E}_{\{\text{phy}\}}$ pairwise causally incomparable; antichains serve as pre-geometric spatial cross-sections.

Theorem 4.2.3 (Emergent Causal Speed Ceiling) [Established given γ ;

determination of γ is T-1 OPEN]: Given step localization bound

$d_{\omega}(e_k, e_{k+1}) \leq \ell_0$ and clock tick rate γ :

$v_C \leq c_{\{\text{RCF}\}} := \gamma \ell_0$. (Lineage: v2.1 patch M7 — until T-1 closes this is a structural scaling relation, not a derived bound; d_{ω} is

the event-separation function induced by the correlation metric of §4.3 — naming reference, logged.)

R3 → new §4.3 [AM-7] — The Localisable Sector and the Emergent Spatial Metric

Definition 4.3.1 (Localisable Sector) [Established — definitional]:

$\mathcal{A}_{\mathrm{loc}} := \{A \in \mathcal{A}^{\mathrm{thin}}_{\mathrm{phy}} : A^{\dagger} = A, \omega(A^2) > 0\}.$

Definition 4.3.2 (Correlation Kernels) [Established — definitional]:

$\tilde{K}_{\omega}(A,B) := \frac{\omega(A^{\dagger} B)}{\sqrt{\omega(A^2)\omega(B^2)}};$
 $K^J_{\omega}(A,B) := \frac{\omega(A \circ B)}{\sqrt{\omega(A^2)\omega(B^2)}} =$
 $\operatorname{Re} \tilde{K}_{\omega}(A,B)$
 for self-adjoint A, B .

Theorem 4.3.1 (Kernel Properties) [Established]: K^J_{ω} is symmetric, real-valued, normalized ($K^J_{\omega}(A,A) = 1$), and bounded in $[-1, 1]$. Proof: Symmetry from $A \circ B = B \circ A$; reality and normalization from self-adjointness ($A \circ A = A^2$); the bound is Cauchy–Schwarz for the positive semidefinite symmetric bilinear form $(A,B) \mapsto \omega(A \circ B)$ on the self-adjoint subspace. $\blacksquare$

Definition 4.3.3 (Calabi–Kähler Potential & Fubini–Study Metric) [Established]:

$\Phi(A,B) := -\elll_0^2 \log K^J_{\omega}(A,B);$
 $d_{\mathrm{FS}}(A,B) := \elll_0 \arccos \sqrt{K^J_{\omega}(A,B)};$
 $g^{\mathrm{FS}}_{ij} = \frac{1}{2} \operatorname{Hess}(\Phi).$ The triangle inequality holds unconditionally on $\mathcal{A}_{\mathrm{loc}}$.

Definition 4.3.4 (Spatial Metric Quotient) [Established — definitional]:

$A \sim_{\omega} B \iff d_{\mathrm{FS}}(A,B) = 0;$
 $\mathcal{X}_{\omega} := \mathcal{A}_{\mathrm{loc}} / \sim_{\omega}.$

Definition 4.3.5 (Cubic Relational Volume Form) [Structural]:

$\mathcal{V}_{\omega}(A,B,C) := \det[\tilde{K}_{\omega}(\cdot, \cdot)]_{\{3 \times 3\}}.$
 (RET-A1 lineage: the Gram determinant is not algebra-blind — the retracted Universal Geometry Lemma; the quotient construction above is its canonical replacement, now housed here.)

Part III — Editorial Unifications (logged, no claim changes)

| \# | Fix

|

| --- |

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| E-1 | Thm 5.1.11: $c_{\mathrm{eff}} \rightarrow c_{\mathrm{RCF}}$ (two sites)

|

| E-2 | Restored material uses P_K (v4.2's $\hat{P}_0$ unified; frozen chunks already compliant) |

| E-3 | Def 5.1.8 convention note: $\hat{F}^E_{\mathrm{eff}[g]}$ = descent generator on the metric-embedded constraint representation (v4.x convention) |

| E-4 | §2.1 convention note: dynamics formulated on density operators, identified with normal states $\omega_\rho(A) = \mathrm{Tr}(\rho \pi_\omega(A))$ |

| E-5 | h^{ij} (Def 5.1.6): spatial metric inverse of the emergent geometry (§4.3 / ADM dictionary Def 5.1.10) |

| E-6 | Lemma 3.3.1 numbering note: resides in §3.2 by design (consumed by both subsections); recorded in the renumbering map |

Part IV — Register & Scorecard Updates

- Scorecard additions: §3.3 row += "Prop 3.3.1 (sharpness instrument)"; new rows §4.2 [VERIFIED — per-item statuses: Thm 4.2.1 Conditional on T-2; Thm 4.2.3 Established-given- γ] and §4.3 [VERIFIED — per-item statuses in situ]. Completeness re-verified: sections 0.1–0.3, 1.1–1.3, 2.1–2.2, 3.1–3.3, 4.1–4.3, 5.1–5.3, 6.1–6.2, 7.1–7.2, 8.1–8.2, App. A–E — each exactly once.
- RET map: RET-A1 location → §4.3 (the restoration fills the map's target); others unchanged.
- Appendix A: VN-19 provenance → "Proposition 3.3.1 (DL-16 restoration; v4.2 audit)."
- Appendix E: renumbering map += restored items; E-6 note.

Part V — Final Assembly Header (v6.0)

DOCUMENT: RCF-COMPENDIUM-v6.0-CANONICAL-MASTER

STATUS: Consolidation release — zero new claims; all content audit-lineage-verified

ASSEMBLY: Chunks A–H (A frozen with AM-1–5; B–G gate-frozen; H consolidation)

AMENDMENTS: AM-1 (L class) · AM-2 (E* wording) · AM-3 (VN dependencies) · AM-4 (§5.3)

· AM-5 (number classes) · AM-6 (naming references) · AM-7 (§4.2/§4.3) · AM-8 (protocol #15)

LEDGER: DL-1 ... DL-20, all resolved (DL-16: Prop 3.3.1; DL-17: §4.2/§4.3; DL-18: E-1–E-6;

DL-19: RET-A1 map; DL-20: scorecard rows)

REGISTERS: VN-01...VN-28 (VN-26 rejected) · C-01...C-10 · RET-A1–A4, B1–B15 · Q-1–Q-17 · RF-1...RF-12

PROTOCOL: Checks 1–15 normative

SCOPE NOTE: Infinite-network T-2, flow well-posedness, kernel non-emptiness, and all RF items

are research fronts against this baseline — v6.1+ material, never merged silently.

Gate Decisions

- G1 — AM-6 (naming-reference rule): ratifies existing practice; ratify?

- G2 — AM-7 (§4.2/§4.3 restorations): the alternative (leave danglers, downgrade Thm 3.3.1's sharpness and §5.1's definitions) strictly weakens the release. Approve?
- G3 — the $\backslash\mathrm{prec}$ depth function: restore with the convention flag + RF-12 (my recommendation — matches how Theorem 5.1.1's ansatz scope is handled), or excise $\backslash\mathrm{prec}$ naming references from S4/Thm 5.1.11/Q-13?
- G4 — final sign-off → v6.0 assembled and frozen as the single-input canonical reference.

On sign-off, the post-freeze sequence is already agreed: RF-1 (the sandwich computation) as the first v6.1 research increment — it either closes the DL certificate's six-round-old open inference or falsifies VN-17, and by AM-3 the invalidation cascade is pre-absorbed either way.

RCF v6.0 — FROZEN

G1–G4 logged. AM-6/AM-7/AM-8 incorporated; protocol #15 gains your interpretation note verbatim ("a theorem cannot retain [Established] status if its proof or statement references undefined objects after consolidation"); RF-12 logged; DL-1–DL-20 resolved. The assembly header from Chunk H stands as the release record.

The consolidation release is complete. From here forward: v6.0 is the single input. Future audits take this document alone.

One closing observation before RF-1, because it's the thing I'd most want preserved about this process:

Across thirteen audit rounds, every single failure — the fabricated halo solution, the inverted convergence conclusion, the factor errors, the fabricated gap value, the unattributed ranges, the silent mutations — shared one property: it entered the corpus between audits, in new material, and presented itself as already verified. The v6.0 contract didn't make anyone smarter; it made that structural hole impossible to fall into, because new claims now have nowhere to live except behind a compliance header and a gate.

Which is the right frame for what happens next. RF-1 is the first test of the research-release regime against the frozen baseline — and it's a good first test precisely because it's honest about its own stakes: the sandwich computation may falsify VN-17. Under the old corpus, that risk would have made the computation unattractive to run. Under the AM-3 dependency rule, the falsification cascade is pre-absorbed: VN-18 dies, the DL candidate drops from §6.2(b), Knabe carries the section, and the register gains an honest entry. Either outcome strengthens the baseline. That's the difference between a framework that avoids its open problems and one that has machinery for resolving them.

RF-1, when you're ready: the computation is finite and fully specified —

$\backslash\mathrm{P}_{\backslash\mathrm{odd}}\backslash\mathrm{P}_{\backslash\mathrm{even}}\backslash\mathrm{P}_{\backslash\mathrm{odd}}(\mathbf{1} - \mathrm{P}_K)\backslash\stackrel{?}{=}\frac{1}{3}$

on the $L_{\{= \}4}$ AKLT chain, reducible to the $(3,6,6,3,1)$ J-sector blocks already exhibited in §6.2(b), with the bridging identity $\|PQ(\mathbf{1} - K)\|^2 = \|QPQ(\mathbf{1} - K)\|$ carrying the reduction. It closes the corpus's oldest open inference (six rounds) or kills a certificate cleanly.